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Review

A Critical Review and Strategic Roadmap of PV Power Forecasting (2016–2026): Addressing Temporal Leakage and Operational Integration Gaps

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Abstract

Photovoltaic (PV) power forecasting plays a central role in power system operation, electricity markets, and the integration of high shares of renewable energy. Over the past decade, forecasting approaches have evolved from classical statistical time-series models to advanced machine learning and deep learning architectures. This review analyses 119 studies published between 2016 and 2026, providing a structured assessment of PV forecasting methodologies, including model types, data requirements, validation strategies, and performance evaluation practices. Beyond summarizing existing approaches, the paper identifies three major methodological gaps in the literature: (i) fragmentation of evaluation metrics, which limits cross-study comparability; (ii) insufficient reporting of data preprocessing procedures and temporal leakage prevention; and (iii) limited integration of forecasting accuracy with economic and operational performance metrics. A systematic comparison of representative studies is conducted to highlight dominant modelling trends and persistent limitations. Beyond a descriptive summary, this review highlights significant limitations in methodological reporting across the 119 studies analysed, particularly regarding temporal leakage prevention in Deep Learning-based forecasting. To address these issues, we introduce a reproducibility checklist and propose a strategic roadmap aimed at strengthening the link between statistical accuracy (e.g., RMSE/MAE) and operational relevance in electricity markets.

Keywords: photovoltaic forecasting; solar power prediction; machine learning; deep learning; forecast evaluation; temporal data leakage; power system integration

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1. Introduction

Photovoltaic (PV) generation has become a central pillar of the global energy transition. The accelerated deployment of solar power plants, driven by decarbonisation commitments and technological cost reductions, is reshaping electricity systems worldwide. However, the increasing penetration of PV generation introduces significant operational challenges related to variability, intermittency, and grid integration. In this context, accurate PV power forecasting is no longer a supplementary analytical tool but an operational prerequisite for modern power systems.

The inherent variability of solar irradiation, influenced by meteorological conditions and geographic factors, directly affects PV output. As PV penetration increases, forecasting accuracy becomes critical for grid stability, reserve scheduling, market participation, and

operational planning. Forecasting errors can lead to imbalance costs, inefficient dispatch decisions, and reduced system reliability. Consequently, significant research efforts have been dedicated to improving forecasting methodologies across different temporal horizons and system scales.

Despite the large volume of literature on PV forecasting, the field remains heterogeneous. Studies differ in terms of application objectives, data sources, forecasting horizons, modelling approaches, evaluation metrics, and reporting practices. This fragmentation complicates objective cross-study comparison and makes it difficult to identify consistent methodological standards. Moreover, although operational applications are frequently discussed, the economic implications of forecasting improvements are often addressed only conceptually, without systematic quantification.

1.1. Aim and Scope of the Review

This review aims to provide a structured and critical synthesis of recent research in PV power forecasting. Rather than focusing exclusively on forecasting algorithms, the article adopts a multi-dimensional perspective that considers the following:

- Application context and operational objectives;
- System scale and deployment characteristics;
- Input data sources and variable selection;
- Forecasting methodologies;
- Preprocessing strategies;
- Forecasting horizons;
- Evaluation metrics;
- Experimental transparency practices.

The scope of the review is therefore not limited to methodological performance comparison. Instead, it seeks to examine how forecasting studies are designed, evaluated, and reported, and how these design choices influence interpretability and practical relevance.

1.2. Contribution of This Review

The contribution of this review lies in its integrative analytical framework. While numerous works summarise forecasting models, fewer studies systematically analyse the structural dimensions of the literature, such as reporting consistency, evaluation standardisation, temporal validation practices, and the linkage between technical accuracy and operational value.

By organising the literature along multiple analytical layers, this review:

- Highlights inconsistencies in evaluation metrics and reporting units;
- Identifies gaps in documentation of data partitioning and leakage prevention practices;
- Examines the alignment between forecasting horizons and operational objectives;
- Discusses the limited quantification of economic impact in existing studies.

This structured approach aims to support improved methodological transparency and facilitate more meaningful comparison across studies.

1.3. Structure of the Paper

The remainder of this article is organised as follows. Section 2 describes the methodology adopted for literature selection and the analytical framework used in this review. Section 3 analyses application contexts and system characteristics. Section 4 discusses input data sources and variable selection practices. Section 5 examines forecasting methods and modelling architectures. Section 6 addresses data processing and feature engineering techniques. Section 7 reviews forecast design and evaluation, including forecasting horizons, temporal resolution, and deterministic and probabilistic metrics. Section 8 discusses

reproducibility and operational relevance, with particular attention to experimental transparency, leakage prevention, and the practical value of forecasting results. Finally, Section 9 presents the main conclusions and outlines future research directions.

2. Methodology of the Review

This article adopts a structured critical review approach, supported by systematic search, screening, and classification procedures. The objective is not to conduct a formal meta-analysis or a protocol-based systematic review in the strict PRISMA sense, but rather to provide a transparent and reproducible mapping of recent PV power forecasting studies according to predefined analytical dimensions. The review, therefore, combines systematic elements, namely database search, explicit inclusion and exclusion criteria, duplicate removal, and structured coding of the selected studies, with a critical narrative synthesis aimed at identifying methodological gaps, reporting inconsistencies, and operational relevance.

This review is guided by the following research question:

- How have PV power forecasting studies published between 2016 and 2026 addressed forecasting methods, validation practices, reproducibility, temporal leakage prevention, and operational relevance?

This research question reflects the objective of the review: not only to classify forecasting model families, but also to assess whether recent PV forecasting research provides sufficient methodological transparency, comparable evaluation practices, and practical relevance for real-world energy system applications.

2.1. Literature Search and Selection Procedure

A systematic review was conducted on scientific publications indexed in the Scopus and ScienceDirect databases covering the period 2016–2026. The search strategy combined keywords including photovoltaic, power generation forecasting, solar energy, and PV power generation, with subject area restrictions to Engineering, Energy, and Multidisciplinary categories. Only journal articles that had reached final publication stage were included. To ensure transparency and allow independent verification of results, all selected articles were open access. The selection process was carried out in three stages.

The initial search yielded 493 articles from journals in the renewable energy field, including Renewable Energy, Applied Energy, Energy, Renewable and Sustainable Energy Reviews, Advances in Applied Energy, IEEE Access, and others.

The first screening phase consisted of reading abstracts and conclusions to ensure that the studies focused specifically on PV power forecasting. This reduced the dataset to 388 articles. After removing duplicates, 369 unique articles remained.

In the second screening stage, stricter inclusion criteria were applied. Only studies that explicitly performed forecasting of photovoltaic electrical power output were retained. Studies forecasting only irradiance or other meteorological variables were excluded. This process resulted in a final corpus of 119 articles, which form the analytical basis of this review.

To improve transparency, the eligibility assessment was based on predefined inclusion and exclusion criteria. Articles were included in the final analysis when they focused on the forecasting of PV electrical power output, presented a clear forecasting model, reported quantitative evaluation metrics such as RMSE, MAE, MAPE, nRMSE, or related indicators, and showed relevance to operational PV applications, including grid operation, scheduling, dispatch, energy management, or power system planning.

Articles were excluded when they addressed broader PV or solar energy topics without a clear PV power forecasting component, predicted only irradiance or meteorological

variables without estimating PV electrical power output, did not present a clearly identifiable forecasting model, did not report standard quantitative evaluation metrics, or lacked sufficient relevance to operational PV applications. Duplicate records were identified using DOI, article title, author names, and publication year, followed by manual verification to avoid counting the same article more than once.

2.2. Descriptive Characterization of the Reviewed Corpus

The final corpus consists of 119 peer-reviewed journal articles published between 2016 and 2026 and selected according to the criteria described in Section 2.1. Before proceeding to the structured analytical assessment, it is useful to provide a brief descriptive overview of the reviewed studies.

Based on the 119 selected studies, research activity is predominantly concentrated in non-tropical regions, representing approximately 83% of the total sample. Tropical regions account for 9%, while the remaining 8% correspond to studies conducted in mixed or multiple geographic locations, still largely dominated by non-tropical contexts. Figure 1 illustrates the geographic distribution of the reviewed studies.

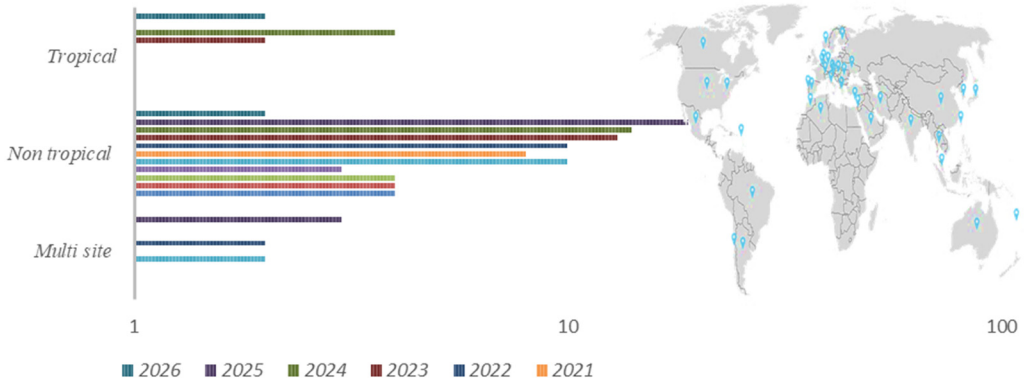


Figure 1. Location of reviewed PV forecasting studies.

This concentration is closely related to countries with large installed PV capacities, such as China and Australia, which also benefit from dense solar radiation monitoring networks and access to consistent satellite data. These conditions facilitate the development and validation of forecasting models using clean and continuous datasets.

In contrast, tropical and high-humidity regions exhibit fundamentally different atmospheric dynamics, characterized by pronounced rainfall variability, frequent convective activity, short-term local cloud formation, and rapid intra-day irradiance fluctuations. These conditions can produce strong and irregular variations in PV output, making short-term forecasting more challenging than in regions with more stable sky conditions. Since the reviewed corpus is predominantly concentrated in non-tropical regions, the forecasting behavior of equatorial and high-humidity climates remains comparatively underrepresented. As a result, models developed and validated mainly under temperate or semi-arid conditions may not generalize well to tropical environments without local adaptation, recalibration, or additional meteorological information.

This geographical imbalance is closely related to a broader data-availability challenge. Countries and regions with mature PV deployment often benefit from dense monitoring networks, long historical production records, and consistent meteorological or satellite datasets. By contrast, newly commissioned PV plants, small installations, and data-scarce locations may lack sufficiently long and representative historical records. This limits the direct applicability of data-intensive machine learning and deep learning models, which generally require large training datasets. Under such conditions, simpler statistical

approaches may be useful as baselines when limited measurements are available, while physical-based models can be particularly relevant because they depend less on historical PV output data and can be applied using system characteristics and meteorological inputs. Hybrid physical–data-driven approaches may also provide a promising direction when limited local data can be combined with physical constraints or transfer-learning strategies.

In addition to geographic scope, operational objectives were identified and classified for each study. The detailed analysis of operational objectives, system scale, climate representation, and other structural dimensions of the literature is presented in Section 3.

2.3. Analytical Framework

To evaluate the 119 studies systematically, this review adopts an analytical framework structured around ten interrelated dimensions.

The first five dimensions relate to application context and input data:

1. Application Context—identification of the operational objectives explicitly mentioned in each study (e.g., grid stability, market participation, energy management).
2. PV Data and System Characteristics—plant capacity, deployment scale, duration of historical data, and temporal resolution.
3. Weather Data and Sensors—meteorological data sources, including local measurements, Numerical Weather Prediction (NWP), satellite data, or combinations thereof.
4. Input Variables—predictor variables used in forecasting models, such as GHI, DNI, DHI, temperature, humidity, wind speed, and historical PV output.
5. System Metadata—technical specifications such as panel orientation, inverter configuration, and shading considerations.
6. The next two dimensions focus on methodological aspects:
7. Model and Architecture—classification of forecasting approaches into statistical, physical-based, machine learning, deep learning, hybrid, and ensemble paradigms.
8. Preprocessing and Feature Engineering—data normalization strategies, missing value handling, outlier removal, and feature selection techniques.
9. The final three dimensions address evaluation and implementation:
10. Evaluation Metrics—error indicators and reporting practices.
11. Reproducibility—availability of code, public datasets, and documentation.
12. Management Relevance—classification of studies according to the level of operational integration:
    - High: explicit measurement of economic or operational impact;
    - Medium: mention of application without quantitative validation;
    - Low: purely technical accuracy evaluation.

This framework enables a structured evaluation of PV forecasting research, bridging technical modelling choices and operational decision relevance.

For dimensions involving categorical classification, additional coding rules were applied to improve consistency across the reviewed studies. System scale and deployment context were classified according to both the physical scale of the PV installation and its operational role, distinguishing between distributed rooftop systems, utility-scale systems, macro or national aggregation, research or small-scale systems, and unclassified studies when the deployment context was not sufficiently specified. Historical data span was grouped according to the length of the available time series.

System metadata were assessed according to whether key technical parameters were explicitly reported, including tilt angle, azimuth, inverter clipping information, degradation rate, shading information, and module or string configuration. Reproducibility was assessed not only in terms of code and dataset availability, but also through methodological transparency indicators, including data splitting strategy, validation strategy, hyperparam-

eter tuning, temporal leakage prevention, and random seed reporting. These coding rules were used to ensure that the ten analytical dimensions were applied consistently across the reviewed corpus.

Figure 2 illustrates the multilayer analysis strategy adopted in this review. The framework combines a qualitative analysis of the full set of 119 studies across ten analytical dimensions with a quantitative benchmarking analysis of the subset of studies reporting both RMSE and MAE. The combined interpretation of these two layers enables a more comprehensive and operationally relevant assessment of PV forecasting research.

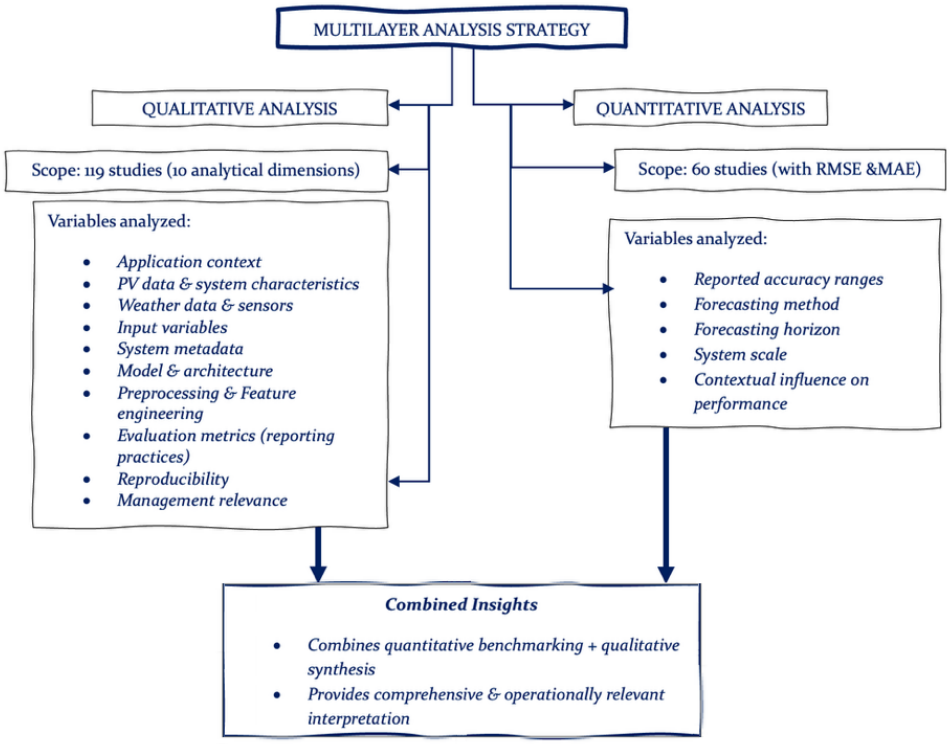


Figure 2. Distribution of forecasting operational objectives.

2.4. Multi-Layer Analysis Strategy

The heterogeneity of evaluation metrics and research focus across PV forecasting studies requires a layered analytical approach. This review applies two complementary analytical layers.

The first layer consists of comparative benchmarking based on 60 studies that consistently report RMSE and MAE metrics. These studies are analysed quantitatively to identify performance variation across forecasting methods, time horizons, and system scales. The objective is not to identify a single "best" model, but to map reported accuracy ranges and contextual influences on performance.

The second layer consists of qualitative synthesis applied to all 119 studies using the ten dimensions of the analytical framework. This layer captures patterns that cannot be evaluated through accuracy metrics alone, including input selection tendencies, validation strategies, dependence on external data, and degree of integration with operational decision processes.

By combining quantitative benchmarking and qualitative structural analysis, the multi-layer strategy ensures that all studies contribute meaningfully to the review, while addressing both performance reporting and implementation feasibility.

3. Application Context and System Characteristics

3.1. Operational Objectives

Operational objectives constitute a central dimension in the reviewed literature, as they define the practical context within which PV power forecasting models are developed and evaluated. While some studies focus primarily on minimizing numerical prediction errors, a substantial portion of the corpus explicitly links forecasting performance to operational decision-making processes.

Based on the systematic classification of the 119 reviewed studies, operational objectives were grouped into three non-mutually exclusive categories: technical, planning, and commercial objectives. Figure 3 summarizes the distribution of these categories across the reviewed corpus.

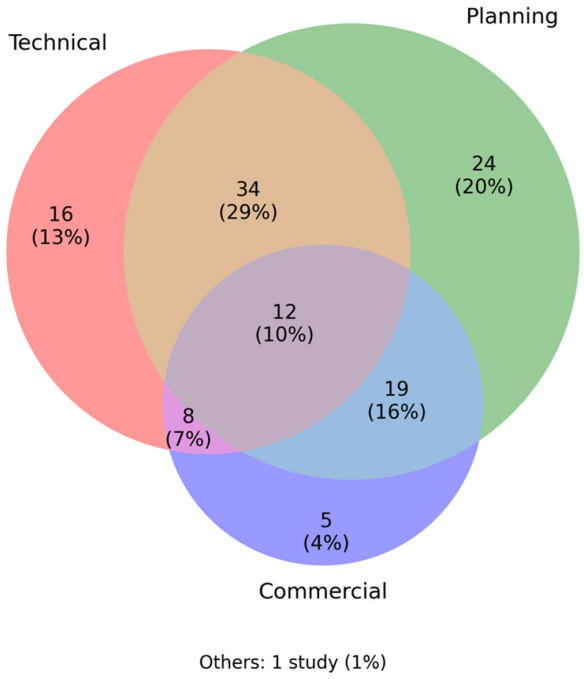


Figure 3. Distribution of operational objectives in the reviewed studies.

Because several studies address more than one objective simultaneously, the categories overlap. The classification therefore reflects the dominant operational framing adopted by each study rather than a strict one-to-one assignment.

3.1.1. Technical Objectives

Technical objectives relate to maintaining system stability and operational security in power systems with increasing PV penetration. In this group, forecasting is positioned as a tool to manage variability and intermittency, support frequency and voltage regulation, and mitigate rapid power fluctuations. Studies focusing on ultra short-term forecasting and nowcasting frequently emphasize ramp-rate management associated with fast-moving cloud cover. Applications in microgrids, smart grids, and virtual power plants are commonly framed within this category, where forecasting contributes to reducing operational risk and preserving grid integrity.

3.1.2. Planning Objectives

Planning objectives concern operational coordination and resource scheduling over short- to medium-term horizons. Forecasts are used to support generation scheduling, dispatch planning, storage management, reserve allocation, and supply–demand balancing.

In this context, forecasting serves as an input to energy management systems at both centralized and distributed scales. Probabilistic forecasting approaches are often associated with planning applications, as they provide uncertainty ranges that facilitate adaptive decision-making under variability.

3.1.3. Commercial Objectives

Commercial objectives frame PV forecasting as an economic requirement linked to electricity market participation. In this category, forecasts are associated with day-ahead bidding, intra-day trading strategies, and compliance with production schedule submission obligations. Forecast accuracy is implicitly connected to the reduction of imbalance penalties and financial risk exposure. Although market participation is frequently cited as a motivation, explicit quantification of economic impact remains limited.

Table 1 maps the identified forecasting roles into their corresponding decision-making contexts, highlighting how technical, planning, and commercial objectives translate into operational functions.

Table 1. Mapping of PV forecasting roles into decision-making contexts.

The Role of PV Forecasting	Relevance to Decision Making
Technical (Stability)	
Reducing PV fluctuations and ramps	Enhancing grid stability
Controlling frequency and voltage	Ensuring system reliability
Planning (Operational)	
Formulating unit commitment and scheduling	Optimizing power dispatching
Managing storage and reserve capacity	Supporting energy management
Commercial (Economic)	
Providing data for market bidding	Supporting trading strategies
Minimizing forecast error/deviations	Avoiding imbalance penalties

The review reveals that operational objectives are frequently stated as motivations for forecasting model development. However, the depth of integration varies substantially across studies. In many cases, operational relevance is described qualitatively, while quantitative validation of system-level or economic impact is not provided. This observation becomes particularly relevant when examining evaluation practices and managerial integration in later sections of this review.

3.2. System Scale and Deployment Context

This subsection groups the reviewed studies according to the size of the PV system analysed and the context in which it is deployed and operated.

System scale refers to the capacity level of the generating unit under examination, ranging from small building-integrated installations to aggregated systems and large centralized power plants.

Deployment context refers to the physical and institutional environment in which the system operates, such as rooftop installations, public facilities, university campuses, regional electricity networks, or grid-connected utility-scale plants.

Based on the reviewed corpus, four main categories were identified: distributed rooftop systems, macro or national aggregation, research or small-scale systems, and utility-scale plants.

Of the 119 reviewed studies, 28% focus on distributed rooftop systems, 30% on utility-scale plants, 17% on aggregated national datasets, and 17% on small-scale research installations.

Based on the reviewed corpus, five categories were considered for system scale and deployment context: distributed rooftop systems, utility-scale systems, macro or national aggregation, research or small-scale systems, and unclassified studies. The unclassified category corresponds to studies for which the system capacity, physical installation, or deployment context was not sufficiently specified. Table 2 summarizes the resulting distribution.

Table 2. Distribution of reviewed studies by system scale and deployment context.

Category	Number of Studies	Percentage
Distributed Rooftop Systems	33	28%
Utility-Scale Systems	36	30%
Macro/National Aggregation	20	17%
Research/Small-Scale Systems	20	17%
Unclassified Studies	10	8%

3.2.1. Distributed Rooftop Systems

This category includes studies focusing on distributed PV installations located on building rooftops and operating as site-specific generation units. Most studies in this group analyse systems installed on commercial buildings, public facilities (e.g., schools, clinics, institutional buildings), or university campuses. Some datasets correspond to a single building, while others cover multiple buildings within the same site.

Deployment contexts include grid-connected rooftop systems, building-integrated PV installations supplying internal demand, and microgrid configurations operating at the building or campus level. These systems are typically characterized by small to medium generation capacity, physical integration with building infrastructure, and operational objectives centered on local energy management.

3.2.2. Macro or National Aggregation

The macro or national aggregation category comprises studies that analyse aggregated PV generation data from multiple locations rather than a single physical installation. These studies operate at city, regional, or national scale and may include hundreds or thousands of distributed residential systems, virtual power plant configurations, or regional datasets treated as a single operational zone under market or grid management arrangements.

Many studies in this group rely on data from national transmission system operators or power system management institutions, with analyses oriented toward large-scale grid stability or market requirements. This category also includes aggregated datasets spanning multiple regions within a country or, in some cases, across several countries. The defining characteristic is the representation of combined PV output at a scale relevant to grid-level operation and system-wide analysis.

3.2.3. Research or Small-Scale Systems

This category includes studies conducted on low-capacity systems located in research facilities, laboratories, pilot plants, or test environments. These installations may consist of a single module or a small number of panels and are often used for methodological validation or technical experimentation.

The primary characteristics of this group are limited generation capacity and experimental deployment contexts that do not fully replicate the operational conditions of commercial power plants. Such studies typically prioritize model development and controlled testing over system-level applicability.

3.2.4. Utility-Scale Systems

The utility-scale category comprises studies focusing on centralized solar power plants with megawatt-level capacity operating as dedicated electricity generation stations. These installations are directly connected to national electricity systems and are often oriented toward grid operation and electricity market participation.

Deployment contexts include ground-mounted solar farms, large power stations, and, in some cases, floating PV systems. Several studies analyse multiple large plants within a single region managed as a unified generation portfolio. The defining characteristics are large generation capacity, centralized operation, and direct interaction with grid and market mechanisms.

3.2.5. Unclassified Studies

A subset of studies does not provide sufficient information regarding system capacity or deployment context and therefore cannot be clearly assigned to one of the four categories. These works are generally methodological or conceptual in nature, focusing on algorithm development, software frameworks, data security, or energy management architectures without reference to a specific physical installation.

3.3. Geographic Distribution and Climate Representation

Geographic and climatic representation is a relevant structural dimension in PV power forecasting research, as forecasting performance is influenced by irradiance variability, seasonal dynamics, and local atmospheric conditions. Based on the reviewed corpus, studies can be grouped into three categories according to spatial scope: single location, multi-site (single country), and multi-site (cross-country) datasets. Each category captures different levels of climatic heterogeneity and spatial diversity.

Of the 119 reviewed studies, 63% rely on single-location datasets, 29% use multi-site data within a single country, and 8% employ cross-country datasets.

3.3.1. Single Location

The single-location category comprises studies based on data collected from a single geographical point, such as a PV plant or a meteorological station. These datasets capture temporal variability at one site and may represent either relatively stable climatic conditions or environments characterized by strong seasonal and daily fluctuations. Depending on the region, this may include temperate climates with distinct seasons, tropical climates with pronounced rainfall variability, or locations exposed to extreme weather events such as heatwaves or dust episodes.

This category emphasizes in-depth observation at a single spatial point, with performance evaluation confined to the climatic characteristics of that specific site.

3.3.2. Multi-Site (Single Country)

The multi-site (single country) category includes studies using data from multiple locations distributed within a single national boundary. This approach captures regional variation, topographic differences, and climatic heterogeneity within a shared regulatory and grid context.

Datasets in this category range from a limited number of stations across different cities to large collections of distributed systems aggregated at regional level. Depending on site selection, studies may represent relatively homogeneous climate zones or incorporate substantial seasonal and meteorological variability across coastal, inland, or climatically distinct regions within the same country.

3.3.3. Multi-Site (Cross-Country)

The multi-site (cross-country) category extends spatial diversity by incorporating datasets from locations situated in different countries. This configuration introduces broader climatic contrasts and allows models to be evaluated under distinct atmospheric regimes, including temperate, semi-arid, desert, or subtropical environments. Cross-country comparisons frequently involve contrasts between temperate and semi-arid climates (e.g., European sites versus Australian desert environments), while high-humidity tropical regions remain comparatively underrepresented.

Although cross-country studies capture greater climatic heterogeneity than single-site or single-country analyses, the reviewed corpus remains predominantly concentrated in non-tropical regions. As a result, representation of equatorial and high-humidity tropical climates is comparatively limited.

The choice of geographic scope directly affects the degree of climatic variability embedded in the dataset and, consequently, the potential generalizability of forecasting models. However, many studies do not explicitly treat climatic heterogeneity as an experimental variable when comparing model performance. This aspect becomes relevant when interpreting evaluation metrics and cross-study comparability in later sections.

3.4. Technical Metadata Reporting Gaps

Beyond spatial distribution, the technical transparency of reported system configurations is equally critical for model benchmarking. In PV forecasting studies, technical metadata describes the physical and operational characteristics of the system under analysis, which directly influence power output and model performance. Relevant metadata typically includes panel inclination (tilt angle), azimuth orientation, inverter clipping, module degradation rate, shading conditions, and module and string configuration.

Across the reviewed corpus, metadata reporting exhibits substantial variability in both completeness and level of detail.

Tilt angle is sometimes reported as precise numerical values, while other studies provide only qualitative descriptions such as “horizontal,” “vertical,” or “optimal.” In some cases, differences in inclination across sensors or sub-arrays within the same plant are documented, indicating internal variability within a single installation.

Azimuth information may be expressed in degrees or simply described using cardinal directions (e.g., south-facing or west-oriented). Some studies consider tracking systems, introducing time-varying orientation during operation.

Inverter clipping is explicitly acknowledged in only a limited subset of studies, typically in cases where DC capacity exceeds nominal inverter power and the effect is incorporated into AC output modelling. Module degradation rate, when reported, is usually expressed as an annual percentage value.

Shading conditions are described with varying levels of precision, ranging from generic acknowledgment of shading presence to more detailed references to inter-row shading or nearby obstructions. Similarly, module and string configuration information varies considerably. While some studies provide complete system specifications including number of panels, installed capacity, electrical configuration, and inverter characteristics others report only aggregate capacity or minimal structural details.

Overall, the degree of technical metadata disclosure differs substantially across studies. Although many forecasting models implicitly depend on system configuration parameters, a significant portion of the literature does not provide comprehensive metadata reporting. This inconsistency limits reproducibility and complicates cross-study comparability, particularly when interpreting reported performance metrics.

Forecast accuracy reported for a fixed-tilt rooftop installation cannot be directly compared with results obtained from a tracking utility-scale plant without accounting for structural differences in system configuration. The omission of key parameters such as tilt angle, orientation, inverter clipping, and shading conditions therefore affects not only transparency but also the methodological validity of cross-study performance comparisons. These observations reinforce the need for more standardized and transparent metadata reporting practices in PV forecasting research.

4. Data Sources and Input Variables

4.1. Meteorological Data Sources

Meteorological inputs constitute a structural component of PV power forecasting models, influencing not only predictive accuracy but also the scalability and transferability of the proposed methods. Based on the reviewed corpus, weather data sources can be classified into five main categories: (i) on-site measurements, (ii) numerical weather prediction (NWP) models, (iii) satellite and reanalysis products, (iv) third-party platforms, and (v) multi-source combinations. In addition, a small subset of studies either does not specify the data source or excludes meteorological variables altogether.

Figure 4 illustrates the conceptual classification of meteorological input categories, distinguishing between single-source and multi-source approaches. Figure 5 presents the empirical distribution observed across the 119 reviewed studies.

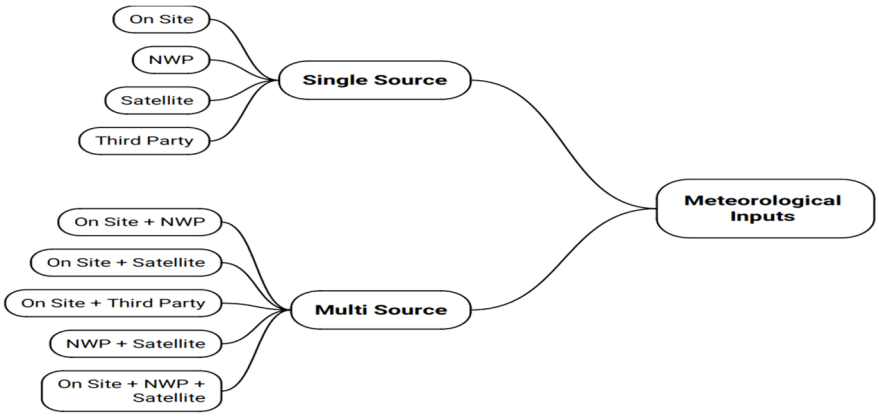


Figure 4. Conceptual classification of meteorological input data sources used in PV power forecasting studies.

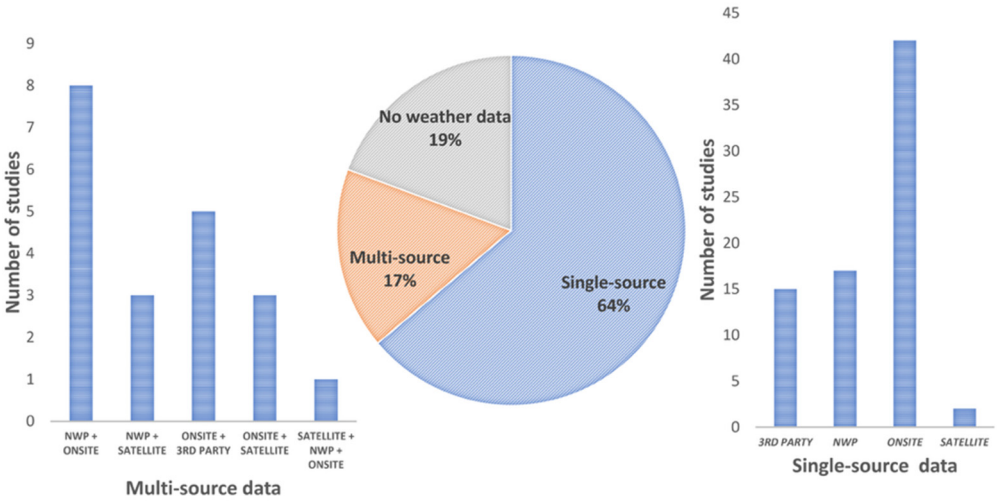


Figure 5. Distribution of meteorological data sources across the 119 reviewed PV forecasting studies.

4.1.1. On-Site Measurements

A substantial share of the literature relies on meteorological variables measured directly at the PV installation or at nearby environmental monitoring stations. Commonly recorded variables include global horizontal irradiance (GHI), direct and diffuse irradiance components, ambient temperature, module temperature, humidity, wind speed, and wind direction.

Data are typically obtained from dedicated on-site weather stations, research platforms (e.g., SunLAB) [1], publicly accessible operational datasets (e.g., DKASC) [2], or nearby meteorological stations located within close proximity to the PV plant (e.g., ESOLMET) [3]. In all cases, the defining feature is the physical measurement of atmospheric variables at or near the site of interest.

4.1.2. Numerical Weather Prediction (NWP) Models

A significant proportion of reviewed studies use NWP outputs as the primary or sole source of meteorological inputs. These data are generated by physics-based atmospheric models operating at regional or global scales.

Frequently cited sources include the European Centre for Medium-Range Weather Forecasts (ECMWF), accessed either through archival systems or through reanalysis datasets. Additional examples include regional or national models such as AROME (Hungarian Meteorological Service) [4], RAMS (Regional Atmospheric Modelling System) [5], WRF-ARW (Advanced Research Weather Research and Forecasting model) [6], and datasets from the Japan Meteorological Agency (e.g., MSM) [7].

Unlike on-site measurements, NWP-based approaches do not require local physical infrastructure. Meteorological variables can be retrieved for any geographic coordinate within the model domain. Nevertheless, forecasting performance becomes dependent on the spatial resolution, update frequency, and bias characteristics of the atmospheric model, which may not fully capture localized microclimatic effects.

4.1.3. Satellite and Reanalysis Products

Satellite-derived and global reanalysis datasets are used as exclusive inputs in a smaller subset of studies. These products are typically provided by international institutions such as NASA, including datasets like the Modern-Era Retrospective Analysis for Research and Applications (MERRA-2) [8].

The defining characteristic of this category is reliance on remote sensing and globally consistent atmospheric datasets. Similar to NWP outputs, satellite-based inputs enable application across broad geographic domains without site-specific infrastructure requirements.

However, effective spatial resolution, cloud detection accuracy, and representation of rapid atmospheric dynamics may influence model performance, particularly in regions characterized by fast-moving cloud formations or strong convective activity.

4.1.4. Third-Party Service Platforms

Several studies retrieve meteorological inputs from third-party platforms through application programming interfaces (APIs). These services aggregate data from satellite products, reanalysis datasets, and NWP forecasts into ready-to-use variables.

Examples identified in the reviewed literature include solar-focused platforms such as Solcast and Solargis, which combine satellite imagery with reanalysis data (often derived from ECMWF ERA5) [9], as well as general weather APIs such as Weatherbit, Dark Sky, Open-Meteo, and visualization platforms such as Ventusky [10]. Observational data accessed through national meteorological agencies via online portals also fall into this category when not directly measured by the research team.

This approach reduces the need for local sensor installation and provides flexible access to both historical and forecast data. However, the preprocessing procedures, bias corrections, and interpolation methods applied by service providers are often not fully transparent, which may limit reproducibility and independent validation.

4.1.5. Multi-Source Integration

An increasing number of studies adopt multi-source strategies that combine two or more meteorological data sources within a single forecasting framework. Typical configurations include on-site measurements combined with NWP outputs, NWP combined with satellite data, or integrated three-source systems.

The rationale behind multi-source integration is to improve completeness and robustness by leveraging complementary strengths: local accuracy from on-site sensors, broader spatial coverage from NWP models, and cloud dynamics information from satellite imagery. While such integration may enhance predictive performance, it increases modelling complexity and data management requirements.

4.1.6. Unspecified Sources and Weather-Excluded Models

A limited number of studies do not clearly report the origin of meteorological variables and are therefore classified as "unspecified." In addition, some works rely exclusively on historical PV power data without incorporating weather variables. These approaches are typically applied to short forecasting horizons and depend strongly on site-specific temporal patterns.

The distribution shown in Figure 5 indicates a predominance of single-source approaches, particularly those based on on-site measurements and NWP data. Multi-source strategies represent a smaller but emerging share of the literature.

The heterogeneity of meteorological data sources described above defines the operational context in which PV forecasting models are developed. Differences in data origin, resolution, and accessibility directly shape the modelling strategies adopted in the literature.

4.2. Input Variable Diversity

In PV power forecasting studies, the selection of input variables is a central determinant of model performance. The literature consistently converges on three primary variable groups, as illustrated in Figure 6.

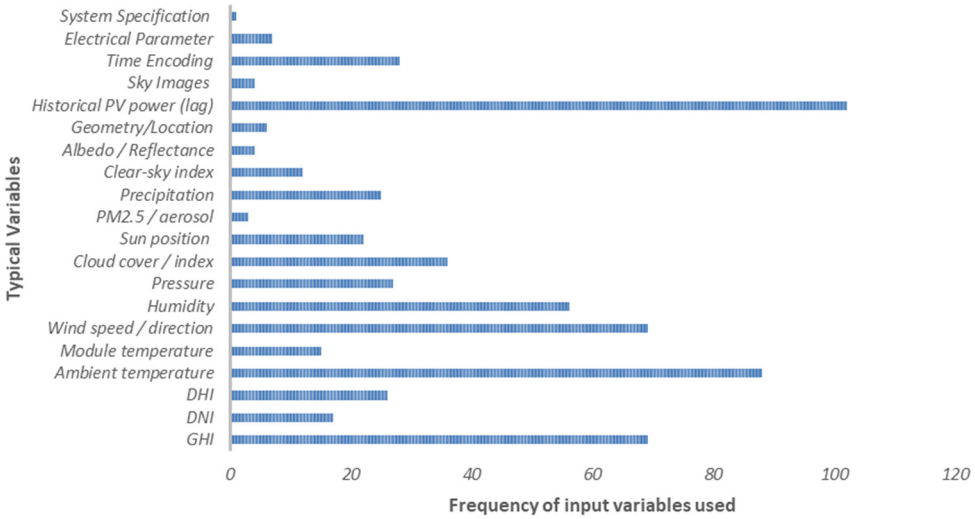


Figure 6. Frequency of input variables used in the 119 reviewed PV power forecasting studies.

First, historical PV power is widely used as the backbone of time-series models, as past generation implicitly captures both meteorological variability and system-specific operational behaviour.

Second, solar radiation parameters, particularly Global Horizontal Irradiance (GHI), constitute the main physical driver of PV output. Direct Normal Irradiance (DNI) and Diffuse Horizontal Irradiance (DHI) are also frequently employed when available.

Third, ambient temperature and wind speed are commonly incorporated to represent thermal effects influencing module efficiency.

Beyond these core variables, many studies include derived parameters computed from time and geographical coordinates. These typically comprise solar position variables (zenith and azimuth angles), Julian day, solar declination, elevation angle, and hour angle. Because such variables can be calculated algorithmically, they enable model enrichment without additional sensing infrastructure.

Some works expand the input space with image-based or operational variables. Sky images are used to monitor cloud movement and sky conditions, while operational parameters, such as system age or inverter efficiency, introduce installation-specific technical characteristics into the forecasting framework.

A smaller subset of studies incorporates more specialised environmental indicators, including air quality metrics (e.g., PM2.5, PM10), aerosols, visibility, atmospheric boundary layer height, and extreme weather variables such as snow depth, hail accumulation, and surface albedo. In a few cases, broader operational metrics, such as plant outage events or grid power consumption, are considered, reflecting an integrated energy management perspective.

Multivariate approaches are common, combining standard meteorological variables with additional parameters such as pressure, cloud cover, clear-sky index, ozone, precipitation, and total column water vapour. In NWP-based frameworks, further atmospheric variables, such as liquid water and ice content and radiation flux components, may be included. In some cases, irradiance components (DNI, DHI) or irradiance on tilted planes are computed from available GHI data through physical modelling rather than direct measurement.

The diversity of input variables observed in Figure 6 is strongly conditioned by the meteorological data sources discussed in Section 4.1, as different sources provide distinct levels of spatial resolution, temporal granularity, and variable availability.

Overall, the diversity of input variables shown in Figure 6 highlights the absence of a universal standard in PV forecasting studies. Variable selection is typically adapted to site characteristics, research objectives, and data availability, reflecting a trade-off between model complexity and predictive performance.

4.3. Scalability vs. Site-Specific Dependency

The selection of data sources in PV forecasting models affects not only predictive accuracy but also their potential for methodological transferability. To assess this dimension, the reviewed studies were evaluated according to two complementary criteria: scalability and site-specific dependency, as summarised in Figure 7. Building upon the source classification presented in Section 4.1 and the variable diversity discussed in Section 4.2, this section examines the structural implications of data selection for model deployment and transferability.

Scalability refers to the ability of a forecasting approach to be applied at new geographical locations without requiring additional hardware installation or exclusive access to private operational records. Site-specific dependency, in contrast, reflects the extent to

which a model relies on data intrinsically tied to a particular PV installation, such as on-site sensors, inverter logs, or long-term historical generation data.

Input Variable Category	Scalability	Site-Dependency
Site-Specific Infrastructure	Low	High
Inverter Log and Global Data	Moderate	High
External Weather Only	High	Low
Historical data	Low	High

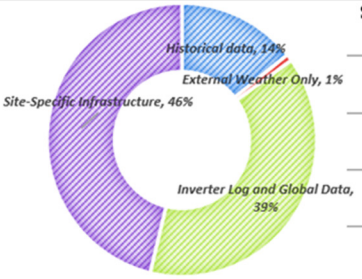


Figure 7. Classification of reviewed PV forecasting studies according to scalability and site-specific dependency.

For consistency, both dimensions were classified into three levels: high, medium, and low (Figure 7).

4.3.1. Scalability

- High scalability: The model can be implemented at a new location using only geographical coordinates, with all required inputs obtained from publicly accessible sources. No permission from system owners or additional hardware is required.
- Moderate scalability: The model does not require new sensor installation but depends on access to inverter or operational logs. Deployment at a new site requires data integration and authorisation.
- Low scalability: Application requires either installation of on-site sensors or the accumulation of extensive historical data before model training becomes feasible.

4.3.2. Site-Specific Dependency

- High dependency: The model relies on data intrinsically linked to a specific PV system, such as inverter records or measurements from on-site instruments. Transferability to other locations requires adaptation or new data acquisition.
- Moderate dependency: The model uses regional weather data (e.g., nearby meteorological stations) without relying on internal system records. This category was not observed in the reviewed studies.
- Low dependency: The model operates independently of local hardware or system-specific data, relying exclusively on regional or global meteorological inputs.

Based on these criteria, the reviewed data sources were grouped into four typologies:

4.3.3. Site-Specific Infrastructure

Approaches in this category require additional sensors (e.g., pyranometers, sky cameras) beyond standard inverter measurements. The need for physical installation at each new site results in low scalability, while reliance on local instrumentation implies high dependency.

4.3.4. Inverter Log and Global Data

These models combine inverter records with satellite or NWP-based weather data. Although no new hardware is required, implementation depends on access to system logs. This leads to medium scalability and high site-specific dependency.

4.3.5. External Weather Only

Models relying exclusively on global meteorological data (without historical PV power) achieve high scalability, as they can be deployed anywhere using publicly available inputs. Their independence from local hardware or operational records results in low site-specific dependency.

4.3.6. Pure Historical Data (Univariate Models)

These models use only historical PV power as input. Because they require long-term site-specific records and cannot be transferred without equivalent data accumulation, they exhibit low scalability and high site-specific dependency.

Overall, as shown in Figure 7, most reviewed studies fall into categories characterised by structural deployment constraints. Site-specific infrastructure represents the largest share (46%), followed by inverter log and global data (39%) and pure historical data (14%). Only 1% of the reviewed studies rely exclusively on external weather data.

When interpreted through the scalability framework, this distribution indicates that 60% of the studies exhibit low scalability, either due to the need for additional hardware installation (46%) or long-term historical data accumulation (14%). Although the operational requirements differ, both conditions prevent immediate transferability to new locations.

This distribution reveals a structural tension within the field: while many studies achieve high predictive accuracy under controlled or instrumented conditions, relatively few propose forecasting frameworks that can be readily deployed across diverse geographic contexts without additional infrastructure or privileged data access.

5. Forecasting Methods and Architectures

The forecasting methodologies reported in the literature have evolved progressively over time, moving from classical statistical formulations toward increasingly data-driven machine learning and deep learning architectures. The following subsections review the main modelling paradigms adopted in PV forecasting studies.

The forecasting approaches reviewed in this section are organized according to their dominant modelling paradigm. It should be noted that these categories are not strictly mutually exclusive. For example, a hybrid architecture may combine statistical, machine learning, or deep learning components, and ensemble frameworks may incorporate models belonging to different methodological families. The classification adopted here therefore reflects the primary modelling strategy emphasized in each study.

Table 3 summarizes the resulting classification, representative architectures, key properties, and number of studies assigned to each model category.

Table 3. Classification and main properties of PV forecasting model categories in the reviewed studies.

Model Category	Subtypes/Architectures	Key Properties/Features	No. of Studies
Statistical	Regularized regression, Time-series (SARIMAX), Bayesian	Linear structure, interpretable, low data requirement, explicit uncertainty (Bayesian), limited nonlinearity	5
Physical-based	Model-chain, Solar-physics-based	Deterministic, based on physical equations, independent of historical data, high interpretability	2
Machine Learning	ANN, MLP, ELM, Tree-based, KNN, Benchmarking	Nonlinear, moderate data requirement, interpretable (tree-based), flexible, suitable for moderate datasets	15

Table 3. *Cont.*

Model Category	Subtypes/Architectures	Key Properties/Features	No. of Studies
Deep Learning	LSTM, GRU, CNN, Transformer, Attention, Spatio-temporal DL	Highly nonlinear, large data requirement, captures complex temporal/spatial dependencies, high accuracy, less interpretable	58
Hybrid	Decomposition, DL + ML, Generative, Input-fusion, Neuro-fuzzy	Combines strengths of multiple different paradigms (e.g., signal processing + ML), robust, handles non-stationarity	29
Ensemble	Stacking, Heterogeneous, Homogeneous, Weighted Average	Aggregates predictions from multiple base models, reduces bias, improves robustness via meta-learners or weighting	10

5.1. Statistical Methods

A subset of the reviewed literature (5 studies) employs classical statistical approaches to model the relationship between input variables and PV power output. These methods rely on explicitly defined mathematical structures and can be grouped into three main categories: (i) regularized regression models, (ii) time-series models, and (iii) Bayesian probabilistic approaches.

5.1.1. Regularized Regression Models

The first category comprises regression-based frameworks enhanced with regularization techniques. Examples include Partial Functional Linear Regression and Reduced Rank Regression [10], which integrate intra-day PV power profiles as functional predictors together with scalar meteorological variables within a unified regression structure.

Regularization mechanisms, such as the Lasso penalty [11], are employed to perform variable selection by shrinking coefficients associated with less informative predictors. This strategy controls model complexity while preserving interpretability of the relationships between inputs and output.

Some of the reviewed approaches are specifically designed to operate without direct irradiance measurements or additional instruments such as pyranometers. In such cases, historical power data and limited climate variables are used to infer system behaviour, enabling application in contexts where measurement infrastructure is constrained.

Overall, this category retains a linear modelling structure but incorporates regularization to mitigate overfitting and improve generalization performance.

5.1.2. Time-Series Models

The second category consists of time-series models, including the Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX), which capture trend and seasonal patterns in historical PV generation data [12]. These approaches are particularly suited to contexts where temporal structure is dominant and the statistical relationship between meteorological conditions and power output is sufficiently stable.

In addition to autoregressive and seasonal components, SARIMAX-type models incorporate exogenous variables (e.g., meteorological inputs) directly into the statistical formulation. Typical preprocessing steps include data decomposition and stationarity testing to ensure compliance with model assumptions.

Within this framework, the contribution of external drivers can be analysed in a transparent and statistically well-defined structure, preserving interpretability while accommodating temporal dependence.

5.1.3. Bayesian Probabilistic Approaches

The third category includes Bayesian probabilistic methods, such as Bayesian Bootstrap Quantile Regression, which focus explicitly on uncertainty quantification. Rather than producing only point forecasts, these models estimate posterior distributions of parameters and generate predictive intervals [13,14].

In this architecture, Bayesian bootstrapping is applied to approximate parameter distributions without imposing strict parametric assumptions. This allows the model to accommodate data variability and provide confidence intervals around forecasts.

Such probabilistic outputs are particularly relevant for grid operators and market participants, who require information on forecast uncertainty and associated risk exposure in addition to expected values.

5.2. Physical-Based Models

Physics-based approaches (2 studies) represent a minority within the reviewed corpus, yet they provide an important methodological contrast to purely data-driven frameworks. Unlike statistical or machine learning models that primarily infer relationships from historical data, physical models explicitly encode the transformation from meteorological inputs to electrical output using deterministic equations derived from solar radiation and PV system physics.

The first study implements a model-chain architecture in which numerical weather prediction (NWP) outputs are sequentially transformed into plane-of-array irradiance, cell temperature, and ultimately electrical power [15]. Multiple sub-model configurations are evaluated to determine the optimal combination of irradiance transposition, temperature modelling, and power conversion steps. A key characteristic of this framework is its independence from historical production data, which enables application to newly commissioned PV plants or sites where long-term operational records are not yet available.

The second study proposes a simplified solar-physics-based architecture that relies on a limited set of meteorological inputs, including cloud cover, ambient temperature, and geographic coordinates [16]. Although historical weather data are used to construct a probabilistic matrix for prediction, the model structure itself remains grounded in physical relationships governing solar radiation and energy conversion. The reduced input requirement facilitates implementation in contexts with limited sensing infrastructure.

Compared to data-driven approaches, physics-based models offer higher interpretability and reduced dependence on extensive training datasets. However, their performance is constrained by the accuracy of the underlying atmospheric inputs and the adequacy of simplified assumptions regarding system behaviour. The small number of such studies in the reviewed literature suggests that, despite their conceptual robustness, physics-based forecasting frameworks are less frequently explored than machine learning-based alternatives.

Nevertheless, these architectures remain particularly relevant for early-stage project development, new installations, and regions where historical production data are scarce, highlighting their complementary role within the broader PV forecasting landscape.

While statistical and physical-based models rely on predefined mathematical structures, machine learning approaches introduce greater flexibility by learning relationships directly from data.

5.3. Machine Learning

A total of 15 reviewed studies employ conventional machine learning (ML) methods for PV power forecasting. Unlike classical statistical models, these approaches relax strict parametric assumptions and are primarily designed to capture non-linear relationships between meteorological inputs and power output. Compared to deep learning architectures,

however, they typically maintain lower computational complexity and reduced structural depth, making them suitable for moderate-sized datasets and operational environments with limited computational resources.

Based on architectural principles, the reviewed ML studies can be grouped into three main categories: (i) neural network-based models, (ii) tree-based methods, and (iii) multi-architecture benchmarking and k-nearest neighbour-based approaches.

5.3.1. Neural Network-Based Models

Five studies rely on feedforward artificial neural network (ANN) structures [17], including Multilayer Perceptron (MLP) [18,19], Extreme Learning Machine (ELM) [20], and Hierarchical ANN configurations [21]. These models are designed to approximate non-linear mappings between weather variables and PV output without incorporating recurrent or deep hierarchical layers.

Several variations are implemented within this group. Some studies apply feature selection techniques, such as sensitivity-based ranking, prior to training the MLP, thereby reducing input dimensionality and improving generalization. Others introduce hierarchical architectures that separate forecasting tasks across temporal scales, for example by first estimating daily output and subsequently refining hourly predictions.

Overall, this category emphasizes relatively shallow neural architectures that balance modelling flexibility with computational efficiency and interpretability.

5.3.2. Tree-Based Methods

Tree-based algorithms represent another important group within conventional ML approaches. These methods construct decision rules through recursive partitioning of the input space and are particularly effective in handling non-linear interactions and mixed-type variables [7,22,23].

Their appeal in PV forecasting lies in their robustness to multicollinearity and their ability to rank variable importance, offering insight into the relative contribution of meteorological drivers [24–26]. In contrast to neural networks, tree-based models provide more explicit structural interpretability, although at the potential cost of reduced smoothness in prediction outputs.

5.3.3. Multi-Architecture Benchmarking and k-Nearest Neighbour (KNN)-Based Approaches

The third group includes studies centred on architecture comparison and pattern similarity methods. Benchmarking studies evaluate multiple ML families simultaneously in order to identify the most suitable architecture for specific forecasting horizons. Examples include comparative assessments combining Kernel Ridge Regression with MLP or Random Forest with Support Vector Regression (SVR) [4,27,28]. Such studies emphasize empirical performance evaluation rather than commitment to a single modelling paradigm.

KNN-based approaches operate on the principle of historical similarity [29]. Forecasts are generated by identifying past instances with meteorological conditions similar to the target scenario and using these neighbours as the predictive basis. Some implementations incorporate multi-level similarity analysis combined with irradiance information to enhance temporal resolution. These methods are particularly effective in environments where historical patterns exhibit strong repetition, even in the absence of explicit physical modelling.

5.3.4. Structural Perspective

Across the reviewed corpus, conventional ML methods occupy an intermediate methodological position. They extend beyond linear statistical models by capturing non-linear relationships, yet they remain less complex than deep learning architectures discussed

in Section 5.4. Their continued relevance reflects a trade-off between predictive flexibility, computational efficiency, and interpretability.

While deep architectures increasingly dominate recent publications, conventional ML methods remain widely applied in contexts with moderate data availability, shorter forecasting horizons, or operational requirements favouring transparent model structures.

More recently, deep learning architectures have become increasingly dominant in PV forecasting research due to their ability to model complex temporal and spatial dependencies in large datasets.

5.4. Deep Learning

A total of 58 studies included in this review employ Deep Learning (DL) architectures, confirming their dominance in recent PV forecasting research. These approaches have become the central methodological paradigm due to their ability to model complex non-linear relationships in renewable energy data without relying on strict distributional assumptions.

DL models are particularly effective in capturing the interactions between meteorological conditions, temporal dynamics, and PV power output. However, this modelling flexibility is accompanied by higher computational requirements and a strong dependence on large training datasets. Based on their architectural design, the reviewed DL studies can be grouped into eight sub-categories.

5.4.1. Recurrent-Based Models

Recurrent neural network architectures represent one of the most widely adopted DL approaches for PV forecasting. These models are designed to capture temporal dependencies in sequential data through gated memory mechanisms that regulate the flow of information across time steps [30].

Among these architectures, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) are commonly used because they mitigate the vanishing gradient problem often encountered in traditional recurrent neural networks. Bidirectional configurations are frequently implemented by arranging two recurrent layers in parallel, allowing the network to process input sequences simultaneously in forward and backward directions [31–33].

Several studies extend these architectures through deeper recurrent structures. Examples include four-layer LSTM models [34], seasonal LSTM architectures with three hidden layers [1], and GRU models with 150 neurons combined with a Multilayer Perceptron (MLP) as a refinement layer [35]. Other studies propose recurrent frameworks designed to manage multiple input variables simultaneously [36] or to generate prediction sequences using Multiple-Input Single-Output (MISO) and Multiple-Input Multiple-Output (MIMO) strategies within LSTM architectures [37,38].

Hybrid recurrent structures are also explored. In some cases, short-term feature extraction using GRU units is combined with LSTM networks in order to capture long-term temporal patterns [39].

Several studies introduce architectural modifications incorporating intermediate targets. Examples include the integration of Extraterrestrial Radiation (ETR) within the M5 framework [40], or the use of Deep Neural Networks (DNN) for feature estimation prior to the main LSTM forecasting model [41]. These designs are often associated with transfer learning strategies aimed at improving generalization across datasets [42].

Error-correction mechanisms are implemented through two-stage architectures such as BC-LSTM and LSTM-Full [43], where auxiliary networks are used to estimate prediction errors or additional variables such as solar position before producing the final forecast [44].

In addition, GRU-based structures are used to address forecasting under multiple weather conditions [45].

5.4.2. Spatio-Temporal Mixed Models

Another important group of DL architectures combines spatial feature extraction with temporal modelling. These spatio-temporal frameworks typically integrate Convolutional Neural Networks (CNN) with recurrent networks.

In these architectures, convolutional layers are placed at the front of the network to extract spatial patterns from meteorological inputs or image-based data [46]. The resulting feature maps are then passed to recurrent units such as LSTM networks [3,47–50] or GRU networks [51,52], which model the temporal evolution of the extracted features.

Some studies further extend this architecture by introducing parallel processing pathways. For example, a concatenation layer may combine outputs from a CNN-LSTM branch with a direct input pathway containing raw power data (often referred to as the power input segment) before producing the final forecast [53].

Other works adopt encoder–decoder architectures composed of two main blocks: an encoder that compresses input sequences into latent feature representations and a decoder that reconstructs the predicted output sequence [8].

Alternative temporal modelling strategies are also explored through Temporal Convolutional Networks (TCN), which employ causal and dilated convolutions to model long-range dependencies while preserving sequence order [54–56]. These networks typically rely on residual blocks to maintain information flow across deep architectures.

Additional feature-weighting mechanisms are introduced through attention modules, which dynamically adjust the importance of input variables, or through Fourier Neural Operators (FNO) that process signals in the frequency domain [57].

Constraint-based and transfer learning approaches are also proposed by introducing parameter restrictions within recurrent cell structures designed to facilitate weight transfer across different data domains [58].

5.4.3. Transformer-Based Models

Transformer-based architectures represent a growing group of DL approaches applied to PV forecasting. These models rely on self-attention mechanisms to capture long-range temporal dependencies in time-series data, avoiding the sequential processing limitations of recurrent architectures [59,60].

In these frameworks, variable selection networks are often incorporated to identify the most relevant input features, while a temporal self-attention decoder is used to learn dependencies across different time steps. Some studies introduce architectural modifications aimed at improving computational efficiency, for example by replacing specific internal components with GRU [61].

Hybrid designs are also explored. Several studies combine the ability of self-attention mechanisms to focus on relevant information with LSTM or Bidirectional Long Short-Term Memory (BiLSTM) decoders in order to capture fine-grained temporal patterns in PV generation data [62,63].

Other approaches introduce cross-attention mechanisms to align short-term predictions with historical observations. Some implementations adapt the shifted-window attention concept to improve computational efficiency when processing moving temporal windows [64].

5.4.4. Attention-Based Models

This category includes models that explicitly integrate attention mechanisms to dynamically weight the relative importance of input variables during the forecasting process.

These architectures allow the model to adaptively distribute attention between short-term fluctuations and long-term temporal patterns, prioritizing different information sources depending on the prevailing weather stability conditions [65].

Several structural variations are implemented to enhance feature extraction. Some studies employ dual-path architectures in which different pathways process data according to time scale or variable type before merging the resulting feature representations in the final prediction layer [66].

Other approaches combine spatial feature extraction using CNN with temporal modelling through recurrent networks such as LSTM or GRU, while attention modules are applied to hierarchically align inter-variable and temporal relationships [67,68].

Data quality enhancement strategies are sometimes integrated directly into the model architecture. In certain studies, outlier detection and cleaning mechanisms are applied before the main prediction stage, whereas other models operate in the frequency domain to decompose seasonal components and long-term trends [9]. In addition, causal convolution layers may be applied before the attention module to capture local temporal context prior to modelling broader correlations in the data [69].

5.4.5. Convolution-Based Models

Another group of studies employs architectures primarily based on convolutional operations. These models transform time-series inputs into feature maps that are processed through stacked convolutional layers, often supported by cross-layer connections designed to preserve information flow and mitigate degradation in deeper networks [70].

Some approaches adopt hierarchical convolutional structures that process data at multiple spatial scales, ranging from local to regional representations, in order to capture spatial dependencies in PV generation patterns [71].

Other architectures employ parallel one-dimensional convolution (1D-CNN) layers with different filter sizes to extract temporal features at multiple resolutions without relying on recurrent memory mechanisms [72].

5.4.6. Structured Deep Learning Models

A smaller group of studies proposes more structured DL architectures that explicitly separate trend and seasonal components in the forecasting process. These models typically rely on shallow stacked networks composed of fully connected layers designed to decompose time-series data into interpretable components [6].

Other approaches extend the N-BEATS (Neural Basis Expansion Analysis for Interpretable Time Series) framework by integrating it with a three-dimensional autoencoder in order to capture richer feature representations [73]. These architectures may also include residual learning components that iteratively refine predictions by correcting errors at successive stages of the training process.

5.4.7. Federated/Deployment Models

A small group of studies explores federated learning architectures designed to process PV time-series data in a distributed manner without transferring raw data to a centralized server. In this paradigm, local models are trained independently at each client node, while only model parameters are shared and aggregated at the central coordinator.

Different architectural implementations are reported. Some studies employ three-layer stacked BiLSTM networks to capture bidirectional temporal dependencies in locally stored datasets [74]. Other approaches use CNNs deployed at each client node to automatically extract relevant weather features from local data streams [75].

To address privacy and security constraints, some architectures integrate GRU with federated optimization algorithms such as FedYogi, combined with differential privacy techniques to ensure confidentiality during model parameter synchronization [76].

Hybrid federated structures are also proposed. In these designs, CNN layers are deployed locally at each client to capture site-specific patterns, while LSTM layers are implemented at the global aggregation level to summarize temporal patterns shared across multiple locations [77].

These distributed architectures are particularly relevant for PV forecasting applications where operational data cannot be centrally shared due to privacy, regulatory, or commercial constraints.

5.4.8. Other Deep Learning Architectures

A small subset of studies proposes hybrid forecasting architectures that combine multiple modelling paradigms within a broader DL framework. Although these models integrate additional techniques, their predictive core remains based on DL architectures.

One example is the GRU-Pool model, which separates linear and non-linear components of the PV generation signal. Linear trends are identified using parameter- and resolution-adaptive algorithms combined with Ordering Points To Identify the Clustering Structure (OPTICS) and Autoregressive Integrated Moving Average (ARIMA) models, while parallel stacks of GRU units are used to capture non-linear dynamics [78].

Another approach introduces the Collaborative Directional Representation (CDR) framework, which combines wavelet-based feature extraction with Rectified Linear Unit (ReLU) activation functions to improve the representation of PV generation patterns [79].

Some architectures integrate Convolutional-GRU (Conv-GRU) video prediction models with diffusion-based techniques and multimodal transformer architectures to enable probabilistic PV forecasting [80].

Other studies adopt horizon-based forecasting strategies, in which different models are applied depending on the prediction horizon. For example, LSTM networks combined with the M5 framework are used for short-term predictions, while transformer-based architectures integrated with the M4 framework are applied for long-term forecasting tasks [81].

Adversarial learning approaches are also explored through Conditional Generative Adversarial Networks (cGAN) built upon CNN and BiLSTM components [2]. In these frameworks, the generator and discriminator networks are trained simultaneously, allowing the model to iteratively improve prediction quality through adversarial learning.

5.5. Hybrid Models

Hybrid models frequently combine components belonging to the methodological families described in Sections 5.1–5.4. For this reason, hybrid architectures are classified here according to their integration strategy rather than as a separate modelling paradigm.

The hybrid modelling category (29 studies) integrates two or more forecasting techniques within a single framework in order to combine complementary strengths. These architectures typically merge signal processing methods, statistical models, machine learning algorithms, or deep learning architectures to improve forecasting accuracy, robustness, or uncertainty representation. Based on their architectural structure, the reviewed hybrid approaches can be grouped into thirteen sub-categories.

5.5.1. Decomposition Models

This category includes architectures in which signal decomposition constitutes the main preprocessing stage. The objective is to separate the original PV generation signal into

components representing different frequency ranges or temporal patterns before applying predictive models.

The Improved Variational Mode Decomposition–Autoregressive Integrated Moving Average–Improved Deep Belief Network (IVMD-ARIMA-IDBN) framework [82] implements a dual-path architecture that separates linear and non-linear components prior to recombining them into the final forecast.

Similarly, the Hybrid Wavelet Grey Correlation–Wavelet Packet Decomposition–Long Short-Term Memory (HWGC-WPD-LSTM) model [83] adopts a multi-branch structure in which several independent LSTM networks process decomposed sub-sequences in parallel before reconstruction.

Several models rely on Variational Mode Decomposition (VMD) as the main signal decomposition technique. The VMD-Convolutional Neural Network (VMD-CNN) architecture combines a two-dimensional Convolutional Neural Network (Conv2D-CNN) pathway with an additional residual branch [84]. The VMD-Extreme Learning Machine (VMD-ELM) framework [85] performs dimensionality reduction followed by a single hidden-layer Extreme Learning Machine (ELM) network. In the VMD-Artificial Neural Network (VMD-ANN) model [86], decomposed components are processed using deep learning backbones such as CNN, LSTM networks, or Residual Networks (ResNet).

Other decomposition-based approaches include transformer architectures that apply self-attention mechanisms to model relationships among decomposed components [87]. The Least Squares Support Vector Regression–Autoregression–Empirical Mode Decomposition (LsSVR-AR-EMD) framework [88] processes each decomposed component independently before merging them into the final output. The Wavelet Packet Decomposition–Long Short-Term Memory (WPD-LSTM) model [89] combines multiple three-layer LSTM networks whose outputs are aggregated through linear weighting. The Wavelet Transform–Deep Convolutional Neural Network (WT-DCNN) architecture [90] employs a dual-path structure consisting of a deterministic branch and a probabilistic branch operating in parallel.

5.5.2. Deep Learning and Machine Learning Probabilistic Models

This category integrates DL architectures with traditional Machine Learning (ML) models in order to generate probabilistic forecasts.

The CNN-BiLSTM-Attention-NGBoost architecture [91] follows a serial pipeline beginning with a one-dimensional Convolutional Neural Network (1D-CNN) for feature extraction, followed by MaxPooling layers, a Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal dependencies, and an attention layer to weight relevant features. The final probabilistic prediction is produced using Natural Gradient Boosting (NGBoost).

The CNN-LSTM-Random Forest (CNN-LSTM-RF) model [92] combines CNN-based spatial feature extraction with sequential temporal modelling through LSTM networks, while Random Forest (RF) acts as the final prediction layer.

5.5.3. Generative Deep Learning Hybrid Models

This category integrates generative models, clustering algorithms, and deep learning predictors. The K-means + Deep Convolutional Generative Adversarial Network (DCGAN) + Quantile Regression Long Short-Term Memory (QRLSTM) framework groups similar weather patterns using K-means clustering, generates synthetic PV power curves through DCGAN, and produces interval forecasts using QRLSTM with a quantile-based loss function [93].

5.5.4. Input-Fusion Hybrid Models

Input-fusion architectures combine multiple data sources within a unified neural network structure. The Hybrid Artificial Neural Network (HANN) model [94] uses a Multi-Layer Feed-Forward Neural Network (MLFFNN) with a single hidden layer to integrate meteorological variables (e.g., air temperature, solar irradiance, and cloud opacity) together with historical PV power data in order to generate 24 h forecasts with 15 min resolution.

5.5.5. Machine Learning—Statistical Hybrid Models

These models combine statistical forecasting techniques with machine learning predictors. The Analog Ensemble (AnEn)—Artificial Neural Network (ANN) framework [5] integrates deterministic forecasts generated by ANN models with probabilistic forecasts derived from Analog Ensemble (AnEn) similarity analysis. The ANN output is used as an additional input for the AnEn model, combining deterministic prediction with probabilistic uncertainty estimation.

Another study applies an ANN with linear regression post-processing, where daily solar irradiance patterns are first grouped using K-means clustering and linear regression is subsequently applied within each cluster to correct prediction biases [95].

5.5.6. Machine Learning—Uncertainty Hybrid Models

This category focuses on hybrid frameworks designed to explicitly quantify prediction uncertainty. One approach employs a two-stage architecture consisting of a RF regression model that generates baseline predictions followed by a conformal prediction layer. The conformal layer integrates k-Nearest Neighbour (KNN) methods with Mondrian binning techniques in order to transform point forecasts into statistically valid confidence intervals [96].

5.5.7. Multi-Stage Hybrid Models

Multi-stage hybrid frameworks apply specialized processing at different stages of the forecasting pipeline. The Fuzzy C-Means—Wavelet Grey Model—Monotonic Composite Quantile Regression Neural Network—Gaussian Approximation (FCM-WGM-MCQRNN-GAQ) architecture [97] integrates Fuzzy C-Means (FCM) clustering for weather classification, Monotonic Composite Quantile Regression Neural Networks (MCQRNN) for non-crossing quantile prediction, and a Gaussian approximation module to reconstruct probabilistic density curves.

Another study proposes a three-stage architecture [98] in which a CNN-GRU-Attention branch processes high-frequency components while Extreme Gradient Boosting (XGBoost) handles low-frequency trends and performs error compensation.

5.5.8. Neuro-Fuzzy Hybrid Models

In this category, hybrid models combine neural learning with fuzzy inference mechanisms. The Adaptive Neuro-Fuzzy Inference System (ANFIS) architecture [99] employs a five-layer structure consisting of fuzzification, rule evaluation, normalization, defuzzification, and aggregation layers. Sugeno-type fuzzy rules with linear output functions are used to integrate neural learning capabilities with fuzzy reasoning.

5.5.9. Optimization Algorithm and Deep Learning Hybrid Models

These architectures integrate optimization algorithms with deep learning predictors. The Improved Particle Swarm Optimization (iPSO)—Clustering—Adaptive LSTM (AdaLSTM) architecture implements an ensemble structure in which input data are grouped using the iPSO algorithm. Each cluster is processed by an LSTM network, and the outputs

are combined through the AdaBoost algorithm to improve prediction accuracy across clusters [100].

5.5.10. Physical Framework Hybrid Models

This category combines physical modelling approaches with machine learning algorithms. The Energy Network Simulation (ENS) Model Chain [101] converts irradiance into electrical power through a physics-based modelling pipeline for each ensemble member or applies an MLP (Multilayer Perceptron) for direct mapping.

The Artificial Intelligence-Based Photovoltaic Forecasting (AI-PVF) framework [102] integrates a physical analytical model with an error-correction layer based on machine learning algorithms adapted to seasonal clusters.

The Physics-Hybrid Artificial Neural Network (PHANN) model [103] combines a two-hidden-layer MLP with a deterministic clear-sky radiation model.

Other studies incorporate Flow-Light Gradient Boosting Machine (Flow-LGBM) models using optical-flow features to estimate cloud motion within a Light Gradient Boosting Machine (LightGBM) framework [104]. In another architecture, a physical solar radiation model is enhanced with two specialized ANN networks to improve direct radiation estimation [105]. The Physics-Constrained Long Short-Term Memory (PC-LSTM) architecture modifies LSTM networks with clipping modules and penalty loss functions in order to ensure that predictions remain consistent with physical constraints [106].

5.5.11. Preprocessing—Deep Learning Hybrid Models

These models focus on improving forecasting stability through data preprocessing techniques. The Locally Weighted Scatterplot Smoothing—Random Forest—Gated Recurrent Unit (LOWESS-RF-GRU) architecture begins with signal smoothing using LOWESS, followed by feature processing through a GRU network enhanced with Repeat Vector layers to expand feature dimensions and TimeDistributed layers to maintain connectivity at each time step [107].

5.5.12. Quantum Deep Learning Hybrid Models

A small number of studies explore the integration of quantum computing with deep neural networks. These approaches combine classical LSTM networks with variational quantum circuits, which use rotation gates and entanglement layers to process data in Hilbert space, potentially enhancing the representation of temporal information in PV forecasting tasks [108].

5.5.13. Signal Processing and Transfer Learning Hybrid Models

The Multi-Dimensional Convolutional Transfer Learning—Multi-Component Integration (MDCTL-MCI) framework [109] employs a three-stage architecture. The Multi-channel Singular Spectrum Analysis (MSSA) unit first performs signal denoising. A Multi-scale Convolutional Interaction (MCI) module then extracts feature interactions across multiple resolutions. Finally, an adaptive fine-tuning stage adjusts the model to new site-specific data through transfer learning, enabling accurate forecasts without retraining the entire network.

5.6. Ensemble Models

Ensemble approaches may incorporate base models from any of the previously described categories. Their classification in this section is therefore based on the strategy used to combine multiple predictors rather than on the underlying model architecture.

The ensemble category (10 studies) integrates predictions from multiple base models in order to improve forecasting performance compared with a single model. By combining

outputs from several predictors, ensemble approaches aim to reduce model bias, improve robustness, and capture complementary information learned by different algorithms.

Several ensemble strategies are identified in the reviewed literature, including stacking methods, heterogeneous ensembles, homogeneous ensembles, and weighted averaging schemes. These approaches combine the outputs of multiple models either through a meta-learner or through predefined weighting mechanisms.

5.6.1. Stacking Models

The stacking approach combines predictions from multiple base models using an additional learning layer that produces the final forecast.

In some studies, four deep learning architectures, including Stacked Long Short-Term Memory (Stacked LSTM) networks and Convolutional Neural Network—Long Short-Term Memory (CNN-LSTM) models, are integrated through Support Vector Regression (SVR) acting as the meta-learner [110,111]. In parallel, traditional machine learning models such as Random Forest Regression (RFR), Gradient Boosting Regression (GBR), and k-Nearest Neighbour (KNN) are combined using linear regression as the final aggregation layer.

Another implementation is the Dynamic Stacking Ensemble—Extreme Gradient Boosting (DSE-XGB) framework [112], which combines predictions from Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) models using Extreme Gradient Boosting (XGBoost) as the meta-learning stage.

Other stacking architectures include dynamic frameworks that integrate Informer, Temporal Convolutional Network (TCN), and Attention-based Gated Recurrent Unit (AttnGRU) models, whose outputs are combined through a decision-tree-based meta-learner [113].

In another configuration, predictions generated by ANN and XGBoost models are further processed using an Extra Trees Regressor, which acts as the final aggregation model to improve predictive accuracy [114].

5.6.2. Heterogeneous Ensembles

Heterogeneous ensemble approaches combine predictions from models belonging to different algorithmic families.

One study implements a heterogeneous ensemble composed of seven distinct machine learning regression models. The individual predictions are aggregated using a normal distribution–based weighting mechanism. Temporal information, including the month and a 30-day historical window, is incorporated into the input features in order to enhance temporal context within the forecasting process [115].

5.6.3. Homogeneous Ensembles

Homogeneous ensemble strategies rely on multiple models of the same type that are trained independently and subsequently combined through an aggregation mechanism.

One example employs a set of MLP networks whose outputs are aggregated using median selection [116]. Another study combines multiple LSTM models with different internal configurations, whose predictions are merged using Quantile Regression Averaging (QRA) [117].

In these approaches, diversity is generated internally through variations in model configuration rather than through the use of different algorithmic families. This strategy improves the stability and robustness of the final forecast.

5.6.4. Weighted Average Ensembles

Weighted averaging approaches determine optimal weights for each model contributing to the ensemble.

The Weight-Varying Combination Forecast Mode (WVCFM) framework [118] combines three different neural networks whose contributions are optimized through a genetic-algorithm-based backpropagation scheme.

Another study determines weights for multiple ANN models using a Mixture Design of Experiments (DOE) approach combined with statistical analysis tools such as Principal Component Analysis (PCA) and Design of Experiments (DOE) techniques [119].

Weighted model combination strategies are often used to mitigate overfitting by balancing the contribution of individual models within the ensemble.

The analysis of forecasting architectures reveals a clear evolution in the literature. Classical statistical and physical-based models remain relevant due to their interpretability and modest data requirements, particularly in data-scarce environments, but their predictive performance is often limited when dealing with highly non-linear PV generation patterns. Machine learning approaches provide greater flexibility and improved accuracy when sufficient historical data are available. More recently, deep learning architectures have become dominant, driven by their ability to capture complex temporal and spatial dependencies in PV data. Hybrid and ensemble strategies further extend these approaches by combining complementary modelling paradigms to enhance robustness and predictive performance. Despite these advances, the diversity of architectures and evaluation practices highlights the absence of standardized methodological frameworks in PV forecasting research.

6. Data Processing and Feature Engineering

Data processing and feature engineering play a critical role in PV forecasting, as they directly influence the quality, consistency, and representativeness of the input data used by forecasting models. The following subsections review the main preprocessing strategies adopted in the literature.

6.1. Normalization and Scaling Strategies

Normalization constitutes a fundamental preprocessing step in PV forecasting, as it directly affects the stability and performance of the forecasting models. Prior to model training, raw data are transformed through several preprocessing operations, including normalization, feature engineering, outlier handling, and data cleaning, with normalization playing a central role in aligning the scale of input variables.

Among the 75 reviewed studies that explicitly apply normalization, six main methods are identified: Min–Max scaling (60% of studies), unspecified methods (23%), Z-score normalization (11%), combined Min–Max and Z-score approaches (2.5%), Max-value scaling (1%), and Median Absolute Deviation-based methods (1%). Range-based techniques such as Min–Max scaling operate by mapping variables to a predefined interval based on their minimum and maximum values, making them particularly suitable for PV data, which are inherently bounded by system capacity. In contrast, Z-score normalization standardizes data using the mean and standard deviation, making it more appropriate when the underlying data distribution is relevant for model training.

Beyond the choice of normalization method, the literature reveals several implementation strategies that influence model performance. These can be grouped into five key dimensions: scope of application, parameter source, target value boundaries, processing stage, and feature grouping.

6.1.1. Scope of Application

This dimension defines whether normalization parameters are applied globally across all data or locally for each site. Three approaches are identified: global, local, and regional normalization. The majority of studies (89%) adopt a local scope, indicating that normal-

ization is performed separately for each PV installation. Global or combined global–local approaches are reported in 7% of the studies, while regional normalization accounts for 4%. Min–Max scaling is predominantly implemented with a local scope (42 out of 45 studies), reflecting the need to adapt scaling parameters to site-specific characteristics such as installed capacity and geographical variability.

6.1.2. Parameter Source

This dimension refers to the dataset used to compute normalization parameters. Three approaches are identified: parameters derived from training data only, from the entire dataset prior to splitting, and incremental approaches. The distribution is nearly balanced, with 49% of studies using training data only and another 49% using the entire dataset, while only 1% employ incremental strategies. Min–Max scaling tends to rely more frequently on training data only (53% of studies), whereas Z-score normalization more often uses the entire dataset (75%).

6.1.3. Target Value Boundaries

This dimension defines the output range of the normalization process. Studies either adopt standard boundaries (e.g., [0, 1]) or define custom ranges tailored to specific applications. Among the reviewed studies, 52% employ custom boundaries, while 48% use standard ranges. For Min–Max scaling, 53% of studies apply custom boundaries, whereas Z-score normalization reports 63%. Customization typically reflects system-specific considerations, such as normalization relative to nominal capacity or adjustments to avoid extreme zero values.

6.2. Feature Engineering and Signal Decomposition

Feature engineering plays a central role in PV forecasting by enabling models to capture temporal patterns and reduce data complexity. The reviewed literature highlights a range of techniques, from simple temporal features to more advanced signal decomposition methods.

6.2.1. Lag Features

Lag features are widely used to capture recurring temporal patterns in PV generation, including intra-day variability (higher production during daylight hours and lower at night) and seasonal effects. A total of 72% of the reviewed studies employ lag features, indicating a strong reliance on historical power data to support forecasting. The remaining 28% of studies rely exclusively on exogenous inputs, such as meteorological variables and time-related features, without incorporating past PV output.

Based on their temporal resolution, lag features can be classified into several categories. Short-term lags (<24 h), representing 15% of the studies, capture recent fluctuations in power output. Daily lags (=24 h), used in 11% of the studies, exploit the periodicity of the diurnal cycle. Multi-day lags (>24 h), reported in 8% of the studies, aim to capture longer-term dependencies, including seasonal trends. In addition, 5% of the studies report the number of lag points without specifying their temporal meaning, while 33% do not provide details on the type of lag used.

6.2.2. Rolling Statistics

Rolling statistics, such as moving averages and rolling standard deviations, are used to smooth highly fluctuating PV data and reveal underlying trends. However, their adoption remains limited, with only 4.2% of the studies incorporating these techniques. Compared with lag features and normalization methods, rolling statistics are not yet widely used, despite their potential to reduce noise and improve short-term pattern extraction.

6.2.3. Signal Decomposition Methods (VMD/EMD)

Signal decomposition techniques are applied to transform complex, non-stationary PV signals into simpler components. Among the reviewed studies, 6.7% employ decomposition methods such as Variational Mode Decomposition (VMD), Improved VMD (IVMD), and Empirical Mode Decomposition (EMD). These approaches are often combined with other techniques, including Empirical Wavelet Transform (EWT), Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) [30], or integrated with dimensionality reduction methods such as Principal Component Analysis (PCA) [31].

6.2.4. Frequency-Domain Methods (Fourier and Wavelet)

Fourier and Wavelet transforms are used to represent time series data in the frequency domain, enabling the identification of hidden cyclical patterns such as daily and seasonal variations. These methods are applied in 5.9% of the studies, representing one of the least adopted preprocessing techniques. Only one study employs pure Fourier analysis [9], whereas most approaches rely on Wavelet-based methods due to their ability to capture both time and frequency information simultaneously. This characteristic makes Wavelet transforms particularly suitable for non-stationary PV data, despite their higher computational complexity.

6.2.5. Time Encoding

Time encoding techniques transform temporal information into numerical features that can be processed by forecasting models. A total of 70 studies apply time encoding, which can be grouped into several categories. The most common approach (41% of studies) uses linear or hour-based encoding (e.g., hour of the day, day of the year, or normalized time indices). Cyclical encoding, employed in 10% of the studies, uses sine and cosine transformations to preserve periodicity. Solar geometry encoding, reported in 13% of the studies, represents time through physically meaningful variables such as solar position angles. Seasonal encoding, which divides data into discrete time blocks, appears in 3% of the studies. The remaining 34% of studies do not specify the encoding method used.

6.2.6. Synthesis

The analysis of 119 studies indicates that lag features and time encoding are the most widely adopted feature engineering techniques, primarily due to their simplicity and effectiveness in capturing temporal patterns in PV generation. In contrast, more advanced approaches, such as signal decomposition and rolling statistics, are less frequently applied, reflecting their higher implementation complexity and computational requirements.

6.3. Outlier Detection and Data Cleaning

Data cleaning and outlier handling are essential preprocessing steps in PV forecasting, as measurement errors, missing data, and non-physical values can significantly affect model performance. A review of 119 studies shows that data handling practices can be grouped into four dominant reporting patterns. These categories reflect how preprocessing procedures are reported in the literature, rather than strictly mutually exclusive methodological classes. Notably, 37% of the studies do not provide transparent reporting on these procedures, highlighting a lack of methodological clarity in the literature.

6.3.1. Comprehensive Data Cleaning

This category includes studies that apply integrated preprocessing procedures addressing multiple data quality issues simultaneously, including missing values, outliers, and operational constraints. These approaches typically combine deletion and imputation techniques within a unified workflow. For example, Isolation Forest is used to detect and

remove outliers, while k-Nearest Neighbour (KNN) methods are applied to fill missing values. Similarly, interquartile range (IQR)-based filtering can be combined with linear interpolation to reconstruct incomplete data series [32].

This category represents the largest share of the reviewed studies (35%), reflecting the need for combined strategies to handle the multiple sources of uncertainty present in PV datasets.

6.3.2. Outlier-Focused Approaches

This category focuses exclusively on the detection and removal of anomalous data points, without explicitly addressing missing values. Outliers may arise from sensor malfunctions, extreme weather events, or maintenance-related disruptions.

The most common approaches include statistical methods such as the three-sigma rule and Z-score thresholding, as well as machine learning techniques including Isolation Forest, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [33], and Hidden Markov Models (HMM) [34]. In addition, simple rule-based filtering is often applied, such as removing values exceeding the panel's rated capacity or eliminating negative power values.

This category accounts for approximately 11% of the reviewed studies.

6.3.3. Physical Filtering

Physical filtering approaches rely on the inherent operational constraints of PV systems and solar geometry. Rather than applying statistical methods, these techniques restrict the dataset to physically meaningful operating conditions.

Typical implementations include filtering data based on solar zenith angle (e.g., >87° or >90°) [4,7], limiting analysis to predefined daylight hours (e.g., 05:00–20:00) [35], or excluding periods with zero solar irradiance. These methods ensure physical consistency in the dataset but do not address missing data or statistical anomalies.

This category represents around 10% of the studies.

6.3.4. Missing Value Handling

This category focuses specifically on the treatment of incomplete data, aiming to preserve time-series continuity. Common techniques include imputation based on historical patterns, such as replacing missing values with the average of the same hour over previous days, using nearest-neighbour interpolation, or removing incomplete records when necessary.

These approaches are reported in approximately 7% of the studies and are typically applied independently of explicit outlier detection strategies.

7. Forecast Design and Evaluation

This section examines how PV forecasting models are designed and evaluated in the literature. Beyond model architecture, forecasting performance depends strongly on the definition of prediction horizons, temporal resolution, and the selection of appropriate evaluation metrics. Particular attention is given to the distinction between deterministic and probabilistic assessment frameworks, as well as to the consistency and transparency of reporting practices, which are essential for meaningful comparison across studies.

7.1. Forecast Horizons and Temporal Design

The evaluation design in PV forecasting is governed by two key temporal dimensions: the forecasting horizon and the temporal resolution of the data. The forecasting horizon defines the prediction timeframe, while temporal resolution determines the data recording interval, directly affecting the level of detail available to the forecasting model.

7.1.1. Forecasting Horizon Classification

The literature reveals significant inconsistencies in how forecasting horizons are defined. Terms such as very short-term and short-term are often used with different time boundaries, which complicates cross-study comparison. To reduce this ambiguity, this review adopts a unified classification framework consisting of four horizon categories.

The very short-term (intra-hour) horizon ($\leq$60 min) is primarily used for real-time applications, including voltage regulation and rapid energy storage response. The short-term (intra-day) horizon (1–6 h) supports local dispatch optimization and coordination of distributed energy resources. The day-ahead horizon (6–48 h) is widely applied in unit commitment, market bidding, and operational planning. Finally, horizons beyond 48 h, extending to several days or weeks, are classified as multi-day or seasonal, and are typically used for maintenance scheduling and long-term generation management.

7.1.2. Temporal Resolution

Similar inconsistencies are observed in the reporting of temporal resolution. Studies describe data intervals in minutes, hours, or combinations of both, often without a consistent classification. In this review, temporal resolution is grouped into four categories based on data recording intervals.

Sub-hourly resolution includes data recorded at intervals below 60 min (e.g., 1, 5, 15, or 30 min), typically used to capture short-term variability. Hourly resolution represents hourly measurements or averages and is the most common standard in energy management, as it effectively captures daily cycles. Multi-resolution approaches are used when multiple time intervals are analysed or combined within the same study. Finally, multi-hour or daily resolution refers to intervals exceeding one hour (e.g., 3 h or daily data), generally used for medium- to long-term trend analysis.

7.1.3. Dominant Horizons and Practical Implications

The reviewed literature shows a clear concentration in specific forecasting horizons. The day-ahead horizon (6–48 h) is the most dominant, accounting for 46% of studies. This reflects the operational requirements of electricity markets, where scheduling and bidding decisions are typically made one day in advance. The very short-term horizon (<1 h) represents the second largest group, accounting for 29% of studies, and is mainly associated with real-time system control.

7.1.4. Resolution–Horizon Consistency

An analysis of 20 studies employing a day-ahead horizon with sub-hourly resolution reveals a lack of methodological justification. In 17 out of 20 studies, the choice of minute-level data is not explicitly justified. Instead, temporal resolution often reflects the format of the available datasets (e.g., DKASC, Elia, AEMO).

Although approximately 70% of these studies acknowledge the trade-off between forecasting accuracy and computational burden, very few quantify whether the use of higher-resolution data leads to statistically significant performance improvements in practical applications.

Several studies, particularly those conducted in Europe and California, note that electricity system operators require outputs at 15 or 30 min intervals. Consequently, even when the forecasting horizon is day-ahead, the output resolution must match these operational requirements.

In addition, ten studies assume that higher-resolution data improves the model's ability to capture short-term variability, such as cloud dynamics, and therefore enhances

forecasting accuracy [6,9,36–43]. However, most of these studies do not perform systematic comparisons between hourly and sub-hourly data to validate this assumption.

7.1.5. Multi-Hour and Multi-Resolution Approaches

Temporal resolutions exceeding one hour are rarely explored. Only one study explicitly employs multi-hour intervals (>60 min), suggesting a general assumption in the literature that finer temporal resolution is inherently superior. This assumption is rarely challenged, despite the potential advantages of coarser resolutions in reducing computational complexity for certain applications, such as large-scale energy storage systems.

Multi-resolution approaches are reported in five studies, which follow two main strategies. First, different resolutions are used to address distinct operational needs, with finer resolutions (15 min to 1 h) supporting real-time decision-making and coarser resolutions (3–24 h) used for planning tasks. Second, multi-resolution frameworks are used to evaluate model robustness across datasets with different temporal granularities.

In most cases, input features, historical data, and output resolution are aligned with the forecasting horizon. However, only two of the five studies provide a clear justification for the selected resolution combinations. While four studies compare performance across resolutions, none explicitly assess whether the observed accuracy improvements justify the additional computational cost.

7.1.6. Methodological Reporting Issues

Significant limitations are also observed in methodological reporting. Among the 15 studies employing intra-day horizons (1–6 h), 73% do not link forecasting performance to operational applications such as redispatch or balancing markets. Only 13% consider practical constraints, such as delays in Numerical Weather Prediction (NWP) updates, despite the fact that input data may be several hours old at the time of forecasting.

Furthermore, 87% of these studies report substantial accuracy degradation when extending the forecasting horizon, with errors increasing by up to a factor of 24 when moving from a 10 min to a 3 h horizon. Notably, 40% of studies recommend shifting to other horizons rather than improving performance within the intra-day range, suggesting that this horizon remains underexplored and insufficiently aligned with operational needs.

Finally, reporting inconsistencies further limit comparability. Four studies do not clearly specify the forecasting horizon, and one study does not report the temporal resolution. Such omissions hinder reproducibility and reinforce the need for more standardized reporting practices in PV forecasting research.

7.2. Deterministic Evaluation Metrics

Deterministic forecasting produces a single predicted value for each future time step, without explicitly quantifying uncertainty. The output consists of point forecasts, which are typically evaluated by measuring the deviation between predicted and observed values. These forecasts are directly applicable to operational decisions, such as unit scheduling and economic dispatch.

The evaluation metrics used in PV forecasting literature can be grouped into six categories according to their methodological role. Absolute error metrics measure the magnitude of the deviation between predicted and actual values in their original units. Relative and percentage metrics express the error as a ratio or percentage of the observed values. Normalized metrics scale the error with respect to installed capacity (e.g., Wp/kWp/MWp) or the mean of the data, enabling comparisons across systems of different sizes. Correlation and fit metrics assess how well the model reproduces the variability and trends of the data, rather than the absolute magnitude of the error. Bias and variance metrics indicate whether the model systematically overestimates or underestimates the actual values. Fi-

nally, skill and comparison metrics evaluate the performance of a model relative to baseline approaches, such as persistence or classical statistical models.

Among these categories, absolute error metrics and correlation metrics are the most consistently reported. The Root Mean Square Error (RMSE) appears in 64% of the reviewed studies, while the Mean Absolute Error (MAE) is reported in 63%, indicating that these two metrics effectively constitute the minimum reporting standard in PV forecasting research. Within the correlation-based metrics, the coefficient of determination (R^2) is used in 39% of the studies, making it the most common non-error indicator.

Other metrics are used less frequently. The Mean Absolute Percentage Error (MAPE) appears in 27% of studies, reflecting its popularity for relative performance interpretation. The Mean Squared Error (MSE) is reported in 16% of studies, typically as an intermediate metric from which RMSE is derived. In contrast, the Mean Bias Error (MBE) appears in only 7% of studies, despite its relevance for identifying systematic over- or underestimation that is not captured by RMSE or MAE.

7.3. Probabilistic Evaluation Metrics and Reporting Issues

Probabilistic forecasting extends deterministic approaches by providing prediction intervals or full probability distributions for each time step, enabling risk assessment and reserve planning in power systems. Among the reviewed literature, only a limited number of studies (9 studies) employ probabilistic forecasting, indicating that this approach remains relatively underexplored compared to deterministic methods.

The evaluation of probabilistic forecasts relies on several types of metrics, which can be grouped according to their function. Probability score metrics, such as the Continuous Ranked Probability Score (CRPS) and Pinball Loss, are used to assess the overall quality of the predicted distribution (5 studies). Interval reliability metrics, including Prediction Interval Coverage Probability (PICP) or coverage probability (CP), evaluate whether the observed values fall within the predicted intervals (4 studies). Interval width metrics, such as Prediction Interval Normalized Average Width (PINAW) or average width (AW), measure the sharpness of the prediction intervals. In addition, some studies report deterministic error metrics (e.g., RMSE, MAE) as reference values for the central tendency of the predicted distribution.

A subset of studies adopts a combined evaluation approach, reporting both deterministic and probabilistic metrics simultaneously. In these cases, the deterministic component is typically evaluated using metrics such as MAE (5 studies), RMSE (4 studies), and MAPE (3 studies), along with normalized metrics (nRMSE, nMAE) and skill scores (SS, FS). The probabilistic component is assessed using metrics such as CRPS [44] and PICP [45], complemented by Pinball Loss [46] and PINAW [47] to evaluate distribution quality and interval sharpness. This combined approach provides a more comprehensive assessment by capturing both prediction accuracy and uncertainty.

Despite these developments, several reporting issues are identified in the literature. In deterministic forecasting, the dominant use of magnitude-based error metrics (RMSE, MAE) often overlooks additional diagnostic information, such as bias (MBE) and skill scores, which are relevant for operational decision-making. In probabilistic forecasting, evaluation practices are frequently incomplete. Some studies report only distribution-based metrics (e.g., CRPS or Pinball Loss) without including interval reliability (PICP) or sharpness (PINAW), while others do the opposite. In addition, several studies continue to rely primarily on deterministic metrics even when probabilistic forecasts are produced, limiting the added value of uncertainty modelling.

Finally, no standardized framework exists for combining deterministic and probabilistic evaluation metrics. The lack of consistency in metric selection results in heterogeneous

reporting practices, making cross-study comparison difficult and limiting the reproducibility and practical interpretation of forecasting performance.

8. Reproducibility and Operational Relevance

This section examines the extent to which PV forecasting studies ensure reproducibility and deliver practical value. In addition to model performance, aspects such as experimental transparency, validation practices, and leakage prevention are critical for reliable comparison and replication. Furthermore, the operational and economic relevance of forecasting results is analysed to assess whether current research effectively supports real-world decision-making in energy systems.

8.1. Experimental Transparency and Validation Practices

Experimental transparency refers to the extent to which the modelling and evaluation procedures are described in sufficient detail to enable replication. Validation practices, in turn, concern the methods used to assess model generalization, including cross-validation and data splitting strategies.

Across the 119 reviewed studies, chronological splitting is the dominant approach, reported in 82% of the cases. In terms of implementation, percentage-based splits (e.g., 80/20 or 70/30) are used in 59 studies, while temporal splits based on specific periods (e.g., day, month, or year) are reported in 35 studies. However, 19 studies (16%) do not clearly specify the data splitting strategy, limiting reproducibility.

Regarding validation practices, only 26% of studies employ cross-validation techniques. These include standard approaches such as 5-fold cross-validation [48] and 10-fold cross-validation [49], as well as time-series-specific methods such as rolling-window validation and nested cross-validation. The remaining 74% rely on a single train/test split, which may not adequately capture temporal variability or model robustness.

Hyperparameter tuning is another critical component of experimental design, yet 44% of studies do not explicitly report the tuning strategy. Among the studies that do, 26% rely on manual tuning, 22% apply grid search, and 8% use Bayesian optimization. When considering the joint use of robust practices, namely chronological splitting, cross-validation, and explicit hyperparameter tuning, only 22% of studies implement all three components simultaneously, indicating limited methodological completeness across the literature.

8.2. Leakage Prevention and Stochastic Reproducibility

Temporal leakage occurs when information that would not be available at the time of prediction is inadvertently used during model training, preprocessing, validation, or testing. In PV power forecasting, this problem is particularly relevant because observations are temporally ordered and strongly autocorrelated. As a result, inappropriate data handling can allow future information to influence the model, leading to overly optimistic performance estimates that may not persist under real-time deployment.

Several forms of temporal leakage can occur in PV forecasting studies. A common case is random shuffling of time-series data before splitting into training and test sets, which allows samples from the future to be used during model development while very similar neighboring samples remain in the test set. Another frequent source of leakage arises when preprocessing operations, such as normalization, standardization, imputation, outlier removal, feature selection, or signal decomposition, are fitted using the full dataset before the train–test split. In such cases, statistical information from the test period indirectly enters the training process. Leakage may also occur when lagged or rolling-window features are constructed without respecting the chronological boundary between training

and testing periods, or when hyperparameters are tuned using information from the final test set.

The main causes of temporal leakage are therefore not necessarily associated with the forecasting algorithm itself, but with the experimental design surrounding the algorithm. Practical mitigation measures include using chronological train–validation–test splits, rolling-origin or walk-forward validation, fitting preprocessing parameters only on the training data, applying imputation and feature selection within the training pipeline, and preserving temporal order when constructing lagged or rolling features. For operational forecasting studies, the test set should represent a future period that was not used during model training, model selection, preprocessing calibration, or hyperparameter tuning.

Against this background, only 37% of the reviewed studies explicitly address leakage prevention. The reported approaches include strict chronological splitting and the restriction of preprocessing operations (e.g., normalization) to the training data only.

In contrast, 63% of studies either do not report leakage prevention strategies or employ approaches that may introduce leakage risk. This omission raises concerns regarding the validity of reported performance metrics, as even minor leakage can lead to overly optimistic results.

Stochastic reproducibility is also rarely addressed. Only 6% of studies report the use of fixed random seeds (e.g., random state = 30 or seed = 0) to ensure consistent results across runs [50]. The absence of such practices limits the ability to reproduce results in models with stochastic components, particularly in machine learning and deep learning frameworks.

Reproducibility is further constrained by limited data availability. Only 8% of studies provide access to datasets through public repositories (e.g., GitHub or dedicated data platforms). Notably, none of the reviewed studies simultaneously report all key reproducibility components, data availability, leakage prevention, and controlled randomness, highlighting a significant gap in experimental transparency.

A detailed analysis of the 119 reviewed papers reveals a consistent lack of methodological transparency regarding validation integrity. Specifically, a majority of the studies do not explicitly report the protocols used to prevent 'temporal leakage' during data shuffling and partitioning. In Deep Learning-based forecasting, this omission may lead to inflated accuracy metrics that fail to persist in real-time deployment. Furthermore, the absence of standardized data-cleaning protocols—particularly regarding the treatment of nocturnal zeros and outlier detection—significantly limits cross-study benchmarking. Taken together, these findings suggest that the current state of PV forecasting has reached a "saturation of models" but a "drought of rigor" in evaluation and reporting practices.". To address these shortcomings, we propose a standardized checklist for future reporting (see Table 4).

Table 4. Recommended reproducibility and rigor checklist for future PV forecasting studies.

Category	Requirement	Rationale
Data Integrity	Explicit reporting of night-time zero handling.	Avoids artificial skewing of error metrics (RMSE/MAE) during low-irradiance periods.
Leakage Control	Use of leakage-safe temporal validation strategies (e.g., chronological split, rolling windows)	Prevents 'Temporal Leakage' (using future data to predict the past).
Benchmarking	Use of relative performance metrics (e.g., skill scores against baseline models such as persistence)	Provides a relative measure of model improvement beyond absolute errors.
Operational	Inclusion of at least one decision-based/economic metric.	Bridges the gap between statistical accuracy and actual grid-reserve costs.
Transparency	Transparent reporting of model configuration, preprocessing steps, and (where possible) code or data availability	Ensures results can be independently verified and reproduced.
Data Integrity	Explicit reporting of night-time zero handling.	Avoids artificial skewing of error metrics (RMSE/MAE) during low-irradiance periods.

8.3. Operational and Managerial Relevance of PV Forecasting

The practical value of PV forecasting lies in its ability to support operational and economic decision-making in energy systems. However, the analysis reveals a substantial gap between claimed applications and quantified impact.

Only 1.7% of studies explicitly quantify real-world benefits, such as economic savings (e.g., €6.82 per month) or improvements in self-consumption (e.g., 12.38%) [14]. In contrast, 95.8% of studies mention potential applications, such as grid stability, dispatch optimization, market bidding, or energy storage management, without providing quantitative evidence of their impact.

A small proportion of studies (2.5%) focus exclusively on technical performance metrics without any discussion of operational relevance. These typically include works centred on methodological improvements, such as privacy-preserving models or probabilistic accuracy, without connection to real-world implementation.

Overall, approximately 98% of studies do not quantify economic or operational value, indicating that the practical contribution of PV forecasting research remains largely unverified. This gap suggests a need for future work to move beyond accuracy metrics and incorporate application-driven evaluation frameworks that explicitly link forecasting performance to decision-making outcomes.

Although improvements in statistical forecasting accuracy are often assumed to translate into operational and economic benefits, this relationship is not universal. A given reduction in RMSE or MAE may result in very different impacts on imbalance costs, reserve activation, storage operation, curtailment, or market revenue depending on the electricity market design, imbalance settlement mechanism, forecast horizon, plant size, bidding strategy, portfolio aggregation, and availability of flexibility resources. Therefore, a direct conversion such as "X% reduction in RMSE corresponds to Y% reduction in imbalance cost" cannot be generalized across studies without a common operational and market framework.

For this reason, future PV forecasting studies should increasingly complement conventional statistical metrics with application-specific indicators. These may include imbalance cost, reserve activation cost, storage dispatch cost, curtailment reduction, revenue deviation, reliability indices, or other decision-oriented indicators. Such metrics are particularly important in market-oriented and grid-support applications, where the economic effect of a forecasting error depends not only on its magnitude, but also on its timing, sign, and operational context.

Table 5 summarizes the practical suitability of the main PV forecasting model families according to data availability, interpretability, complexity, and operational use case.

Table 5. Practical engineering suitability of PV forecasting model families.

Model Family	Practical Engineering Suitability
Physical models	Suitable for new PV plants or data-scarce sites; interpretable and less dependent on historical PV data, but sensitive to weather inputs and system assumptions.
Statistical models	Suitable as transparent baselines and for stable systems with limited data; less effective under highly nonlinear or rapidly variable conditions.
Machine learning models	Suitable when structured historical and meteorological data are available; good balance between accuracy, complexity, and interpretability.
Deep learning models	Suitable for data-rich sites and high-frequency forecasting; require careful validation to avoid temporal leakage, overfitting, and poor transferability.
Hybrid and ensemble models	Suitable for complex operational contexts where robustness is important; often accurate, but more difficult to reproduce and interpret.

8.4. Strategic Roadmap for Future PV Forecasting Research

To make the future outlook more actionable, Table 6 summarizes the proposed short-, medium-, and long-term roadmap for PV forecasting research.

Table 6. Staged roadmap for future PV power forecasting research.

Stage	Core Goals	Key Technologies and Priorities	Expected Milestones
Short term	Improve transparency, comparability, and reproducibility of PV forecasting studies.	Standardized reporting of data preprocessing, chronological validation, leakage prevention, baseline models, error metrics, and reproducibility information.	Wider use of reproducibility checklists, clearer reporting of train–validation–test splits, consistent treatment of nocturnal data and outliers, and systematic comparison against simple baselines.
Medium term	Strengthen methodological robustness and operational interpretation of forecasting models.	Relative performance metrics, probabilistic forecasting, physics-informed machine learning, hybrid physical–data-driven models, and clearer justification of model complexity.	Broader adoption of skill scores, uncertainty-aware evaluation, benchmark datasets, ablation studies, and model selection rules based on data availability and engineering context.
Long term	Integrate PV forecasting into real-world operational and economic decision-making.	Forecast-driven market bidding, reserve scheduling, storage dispatch, curtailment management, grid-support services, and digital-twin or co-simulation environments.	Routine evaluation of forecasts using operational value metrics such as imbalance cost, reserve activation cost, storage dispatch cost, revenue deviation, and reliability-related indicators.

9. Conclusions and Future Research Directions

This review analysed 119 studies on photovoltaic (PV) power forecasting published between 2016 and 2026, with the objective of identifying dominant methodological trends, persistent evaluation gaps, and the degree of alignment between forecasting research and operational needs. The analysis shows that PV forecasting has evolved substantially over the last decade, moving from classical statistical and physical-based approaches toward increasingly complex machine learning, deep learning, hybrid, and ensemble architectures.

Classical statistical and physical-based models remain relevant, particularly in contexts where interpretability, limited data availability, or low computational requirements are important. Machine learning models provide greater flexibility and are widely used when structured historical and meteorological data are available. Deep learning architectures have become dominant in the recent literature, especially for short-term and very short-term forecasting, due to their ability to capture nonlinear temporal patterns in large datasets. Hybrid and ensemble models further extend these capabilities by combining complementary modelling strategies, improving robustness under non-stationary conditions and, in some cases, supporting probabilistic forecasting.

However, the review also shows that progress in model complexity has not always been accompanied by equivalent progress in methodological transparency and operational validation. A first major gap concerns the fragmentation of evaluation metrics. RMSE and MAE remain the most widely used indicators, but additional metrics are reported with inconsistent definitions, normalization procedures, and units. This limits objective comparison across studies and makes it difficult to determine whether reported performance improvements are genuinely meaningful.

A second important limitation concerns data preprocessing and experimental design. Many studies do not provide sufficient information on normalization procedures, feature engineering, outlier handling, hyperparameter tuning, validation strategy, or temporal leakage prevention. This is particularly relevant for machine learning and deep learning

models, where inappropriate data splitting or preprocessing can produce overly optimistic results. The lack of reporting on random seeds, code availability, and dataset accessibility further limits reproducibility.

A third gap relates to the increasing complexity of forecasting architectures. Deep learning, hybrid, and ensemble models often involve multi-stage designs, signal decomposition, attention mechanisms, or combinations of several predictors. While these approaches may improve accuracy, their additional complexity is not always justified through systematic comparisons with simpler baselines. As a result, it is sometimes unclear whether performance gains arise from genuinely superior modelling capability or from dataset-specific tuning and experimental design choices.

A fourth limitation concerns the treatment of temporal design. Forecasting horizons and temporal resolutions are not consistently defined across the literature, and their selection is often driven by data availability rather than by operational requirements. More systematic analysis is needed to clarify the trade-off between temporal resolution, forecast accuracy, computational burden, and practical value for applications such as real-time control, market bidding, reserve scheduling, and energy management.

Finally, the review highlights a weak connection between statistical forecasting performance and operational or economic value. Although most studies mention applications such as grid stability, market participation, or energy management, only a small fraction explicitly quantifies how forecasting improvements affect reserve requirements, imbalance costs, storage operation, curtailment, or revenue deviations. This indicates that much of the literature remains focused on numerical accuracy rather than decision-oriented performance.

Based on these findings, this review proposes a transition toward a more operationally relevant PV forecasting paradigm, referred to here as Forecasting 2.0, structured around three strategic axes.

First, the rigor gap should be addressed by improving transparency in data handling, validation design, and temporal leakage prevention. Future research should clearly report how training, validation, and test sets are constructed, ensure that preprocessing parameters are derived only from appropriate data subsets, and provide sufficient information to allow reproducibility. Physics-informed machine learning may also play an important role by incorporating solar geometry, clear-sky models, or physical constraints into data-driven architectures, improving plausibility under extreme or poorly represented conditions.

Second, the evaluation gap should be reduced through more consistent benchmarking practices. Conventional error metrics such as RMSE and MAE should be complemented by relative performance indicators, such as skill scores against appropriate baselines, and by probabilistic metrics when uncertainty forecasts are produced. The reproducibility checklist proposed in this review is intended to support more consistent reporting of nocturnal data treatment, outlier handling, normalization, validation strategy, random seeds, and data or code availability where feasible.

Third, the integration gap should be addressed by linking forecasting evaluation more directly to operational value. Future studies should progressively include application-oriented indicators such as imbalance cost, reserve activation cost, storage dispatch cost, curtailment reduction, revenue deviation, or reliability-related metrics. This does not imply that a universal conversion between RMSE reduction and economic benefit exists, since such a relationship depends on market design, plant size, forecast horizon, bidding strategy, and flexibility resources. Rather, it means that forecasting models should increasingly be evaluated within the operational contexts in which they are intended to be used.

Overall, PV forecasting research has achieved significant methodological progress, but further advances require more than increasingly complex models. The field now

needs stronger reporting standards, more robust validation practices, clearer justification of model complexity, and closer integration between forecasting accuracy and real-world decision-making. Addressing these issues will help ensure that PV forecasting becomes not only more accurate, but also more transparent, comparable, reproducible, and operationally relevant.

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References

1. Campos, F.D.; Sousa, T.C.; Barbosa, R.S. Short-Term Forecast of Photovoltaic Solar Energy Production Using LSTM. *Energies* **2024**, *17*, 2582. [CrossRef]
2. Huang, X.; Li, Q.; Tai, Y.; Chen, Z.; Liu, J.; Shi, J.; Liu, W. Time series forecasting for hourly photovoltaic power using conditional generative adversarial network and Bi-LSTM. *Energy* **2022**, *246*, 123403. [CrossRef]
3. Tovar, M.; Robles, M.; Rashid, F. PV power prediction, using CNN-LSTM hybrid neural network model. Case of study: Temixco-Morelos, México. *Energies* **2020**, *13*, 6512. [CrossRef]
4. Markovics, D.; Mayer, M.J. Comparison of machine learning methods for photovoltaic power forecasting based on numerical weather prediction. *Renew. Sustain. Energy Rev.* **2022**, *161*, 112364. [CrossRef]
5. Cervone, G.; Clemente-Harding, L.; Alessandrini, S.; Monache, L.D. Short-term photovoltaic power forecasting using Artificial Neural Networks and an Analog Ensemble. *Renew. Energy* **2017**, *108*, 274–286. [CrossRef]
6. Jannah, N.; Gunawan, T.S.; Yusoff, S.H.; Hanifah, M.S.A.; Sapihie, S.N.M.; Adlan, H. Fast and Accurate Solar Power Generation Forecasting Using Advanced Deep Learning: A Novel Neural Basis Expansion Analysis Framework. *IEEE Access* **2025**, *13*, 193414–193429. [CrossRef]
7. Persson, C.; Bacher, P.; Shiga, T.; Madsen, H. Multi-site solar power forecasting using gradient boosted regression trees. *Sol. Energy* **2017**, *150*, 423–436. [CrossRef]
8. Ibrahim, M.S.; Gharghory, S.M.; Kamal, H.A. A hybrid model of CNN and LSTM autoencoder-based short-term PV power generation forecasting. *Electr. Eng.* **2024**, *106*, 4239–4255. [CrossRef]
9. Yuan, L.; Wang, X.; Sun, Y.; Liu, X.; Dong, Z.Y. Multistep photovoltaic power forecasting based on multi-timescale fluctuation aggregation attention mechanism and contrastive learning. *Int. J. Electr. Power Energy Syst.* **2025**, *164*, 110389. [CrossRef]
10. Rosso, A.P.; Rampinelli, G.A.; Schaeffer, L. Experimental development of a method of short and medium-term photovoltaic generation forecasting using multivariate statistics and mathematical modeling. *Energy Rep.* **2024**, *12*, 1710–1722. [CrossRef]
11. Wang, G.; Su, Y.; Shu, L. One-day-ahead daily power forecasting of photovoltaic systems based on partial functional linear regression models. *Renew. Energy* **2016**, *96*, 469–478. [CrossRef]
12. Benitez, I.B.; Ibañez, J.A.; Lumabad, C.I.I.I.D.; Cañete, J.M.; Principe, J.A. Day-Ahead Hourly Solar Photovoltaic Output Forecasting Using SARIMAX, Long Short-Term Memory, and Extreme Gradient Boosting: Case of the Philippines. *Energies* **2023**, *16*, 7823. [CrossRef]
13. Bozorg, M.; Bracale, A.; Caramia, P.; Carpinelli, G.; Carpita, M.; De Falco, P. Bayesian bootstrap quantile regression for probabilistic photovoltaic power forecasting. *Prot. Control Mod. Power Syst.* **2020**, *5*, 21. [CrossRef]

14. Roldán-Blay, C.; Abad-Rodríguez, M.F.; Abad-Giner, V.; Serrano-Guerrero, X. Interval-based solar photovoltaic energy predictions: A single-parameter approach with direct radiation focus. *Renew. Energy* **2024**, *230*, 120821. [CrossRef]
15. Gandoman, F.H.; Aleem, S.H.E.A.; Omar, N.; Ahmadi, A.; Alenezi, F.Q. Short-term solar power forecasting considering cloud coverage and ambient temperature variation effects. *Renew. Energy* **2018**, *123*, 793–805. [CrossRef]
16. Mayer, M.J.; Gróf, G. Extensive comparison of physical models for photovoltaic power forecasting. *Appl. Energy* **2021**, *283*, 116239. [CrossRef]
17. Nicoletti, F.; Bevilacqua, P. Hourly Photovoltaic Production Prediction Using Numerical Weather Data and Neural Networks for Solar Energy Decision Support. *Energies* **2024**, *17*, 466. [CrossRef]
18. Nespoli, A.; Ogliari, E.; Leva, S.; Massi Pavan, A.; Mellit, A.; Lughi, V.; Dolara, A. Day-ahead photovoltaic forecasting: A comparison of the most effective techniques. *Energies* **2019**, *12*, 1621. [CrossRef]
19. Brester, C.; Kallio-Myers, V.; Lindfors, A.V.; Kolehmainen, M.; Niska, H. Evaluating neural network models in site-specific solar PV forecasting using numerical weather prediction data and weather observations. *Renew. Energy* **2023**, *207*, 266–274. [CrossRef]
20. Hossain, M.; Mekhilef, S.; Danesh, M.; Olatomiwa, L.; Shamshirband, S. Application of extreme learning machine for short term output power forecasting of three grid-connected PV systems. *J. Clean. Prod.* **2017**, *167*, 395–405. [CrossRef]
21. Li, Z.; Rahman, S.M.M.; Vega, R.; Dong, B. A hierarchical approach using machine learning methods in solar photovoltaic energy production forecasting. *Energies* **2016**, *9*, 55. [CrossRef]
22. Jain, A.; Gupta, R.K.; Kumar, M. Solar Power Forecasting Using Machine Learning Techniques. In *Reshaping Environmental Science Through Machine Learning and IoT*; Gupta, R., Jain, A., Wang, J., Pateriya, R., Eds.; IGI Global Scientific Publishing: Hershey, PA, USA, 2024; pp. 324–336. [CrossRef]
23. Grzebyk, D.; Alcañiz, A.; Donker, J.C.B.; Zeman, M.; Ziar, H.; Isabella, O. Individual yield nowcasting for residential PV systems. *Sol. Energy* **2023**, *251*, 325–336. [CrossRef]
24. Nikolaidis, P. AI-Enhanced Photovoltaic Power Prediction Under Cross-Continental Dust Events and Air Composition Variability in the Mediterranean Region. *Energies* **2025**, *18*, 3731. [CrossRef]
25. Abdullah, B.U.D.; Khanday, S.A.; Islam, N.U.; Lata, S.; Fatima, H.; Nengroo, S.H. Comparative Analysis Using Multiple Regression Models for Forecasting Photovoltaic Power Generation. *Energies* **2024**, *17*, 1564. [CrossRef]
26. Scott, C.; Ahsan, M.; Albarbar, A. Machine learning for forecasting a photovoltaic (PV) generation system. *Energy* **2022**, *278*, 127807. [CrossRef]
27. Tucci, M.; Piazzi, A.; Thomopulos, D. Machine Learning Models for Regional Photovoltaic Power Generation Forecasting with Limited Plant-Specific Data. *Energies* **2024**, *17*, 2346. [CrossRef]
28. Sulaiman, M.H.; Jadin, M.S.; Mustaffa, Z.; Azlan, M.N.M.; Daniyal, H. Short-Term forecasting of floating photovoltaic power generation using machine learning models. *Clean. Energy Syst.* **2024**, *9*, 100137. [CrossRef]
29. Sangrody, H.; Zhou, N.; Zhang, Z. Similarity-Based Models for Day-Ahead Solar PV Generation Forecasting. *IEEE Access* **2020**, *8*, 104469–104478. [CrossRef]
30. Elsaraiti, M.; Merabet, A. Solar Power Forecasting Using Deep Learning Techniques. *IEEE Access* **2022**, *10*, 31692–31698. [CrossRef]
31. Sharadga, H.; Hajimirza, S.; Balog, R.S. Time series forecasting of solar power generation for large-scale photovoltaic plants. *Renew. Energy* **2020**, *150*, 797–807. [CrossRef]
32. Mellit, A.; Pavan, A.M.; Lughi, V. Deep learning neural networks for short-term photovoltaic power forecasting. *Renew. Energy* **2021**, *172*, 276–288. [CrossRef]
33. Mansour, A.A.; Tilioua, A.; Touzani, M. Bi-LSTM, GRU and 1D-CNN models for short-term photovoltaic panel efficiency forecasting case amorphous silicon grid-connected PV system. *Results Eng.* **2024**, *21*, 101886. [CrossRef]
34. Dhaked, D.K.; Dadhich, S.; Birla, D. Power output forecasting of solar photovoltaic plant using LSTM. *Green Energy Intell. Transp.* **2023**, *2*, 100113. [CrossRef]
35. Castillo-Rojas, W.; Bekios-Calfa, J.; Hernández, C. Daily Prediction Model of Photovoltaic Power Generation Using a Hybrid Architecture of Recurrent Neural Networks and Shallow Neural Networks. *Int. J. Photoenergy* **2023**, *2023*, 2592405. [CrossRef]
36. Hayawi, K.; Maliakkal, H.; Venugopal, N.; Hussain, T.M.; Rajagopalan, G.B. Machine learning—Driven solar forecasting in dust-prone regions for sustainable energy systems. *Sol. Energy Adv.* **2025**, *5*, 100108. [CrossRef]
37. Delgado, C.J.; Alfaro-Mejía, E.; Manian, V.; O'Neill-Carrillo, E.; Andrade, F. Photovoltaic Power Generation Forecasting with Hidden Markov Model and Long Short-Term Memory in MISO and SISO Configurations. *Energies* **2024**, *17*, 668. [CrossRef]
38. Nelega, R.; Greu, D.I.; Jecan, E.; Rednic, V.; Zamfirescu, C.; Puschita, E. Prediction of Power Generation of a Photovoltaic Power Plant Based on Neural Networks. *IEEE Access* **2023**, *11*, 20713–20724. [CrossRef]
39. Radzi, P.N.L.M.; Mekhilef, S.; Shah, N.M.; Akhter, M.N.; Seyedmahmoudian, M.; Almalaq, Y.; Wahyudie, A. Enhancing short-term solar power forecasting using a hybrid deep learning approach and a comparative analysis of machine learning models. *Energy Convers. Manag. X* **2026**, *29*, 101431. [CrossRef]
40. Jo, S.C.; Jin, Y.G.; Yoon, Y.T.; Kim, H.C. Methods for integrating extraterrestrial radiation into neural network models for day-ahead pv generation forecasting. *Energies* **2021**, *14*, 2601. [CrossRef]

41. Wai, R.J.; Lai, P.X. Design of Intelligent Solar PV Power Generation Forecasting Mechanism Combined with Weather Information under Lack of Real-Time Power Generation Data. *Energies* **2022**, *15*, 3838. [CrossRef]
42. Bellagarda, A.; Grassi, D.; Aliberti, A.; Bottaccioli, L.; Macii, A.; Patti, E. Effectiveness of neural networks and transfer learning to forecast photovoltaic power production. *Appl. Soft Comput.* **2023**, *149*, 110988. [CrossRef]
43. Zhang, J.; Verschae, R.; Nobuhara, S.; Lalonde, J.F. Deep photovoltaic nowcasting. *Sol. Energy* **2018**, *176*, 267–276. [CrossRef]
44. Tao, C.; Lu, J.; Lang, J.; Peng, X.; Cheng, K.; Duan, S. Short-term forecasting of photovoltaic power generation based on feature selection and bias compensation–lstm network. *Energies* **2021**, *14*, 3086. [CrossRef]
45. Kuo, W.C.; Chen, C.H.; Chen, S.Y.; Wang, C.C. Deep Learning Neural Networks for Short-Term PV Power Forecasting via Sky Image Method. *Energies* **2022**, *15*, 4779. [CrossRef]
46. Wang, K.; Qi, X.; Liu, H. A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network. *Appl. Energy* **2019**, *251*, 113315. [CrossRef]
47. Agga, A.; Abbou, A.; Labbadi, M.; El Houm, Y.; Ali, I.H.O. CNN-LSTM: An efficient hybrid deep learning architecture for predicting short-term photovoltaic power production. *Electr. Power Syst. Res.* **2022**, *208*, 107908. [CrossRef]
48. Jakoplić, A.; Franković, D.; Havelka, J.; Bulat, H. Short-Term Photovoltaic Power Plant Output Forecasting Using Sky Images and Deep Learning. *Energies* **2023**, *16*, 5428. [CrossRef]
49. Li, G.; Xie, S.; Wang, B.; Xin, J.; Li, Y.; Du, S. Photovoltaic Power Forecasting with a Hybrid Deep Learning Approach. *IEEE Access* **2020**, *8*, 175871–175880. [CrossRef]
50. Yu, D.; Lee, S.; Lee, S.; Choi, W.; Liu, L. Forecasting photovoltaic power generation using satellite images. *Energies* **2020**, *13*, 6603. [CrossRef]
51. Thipwangmek, N.; Suetrong, N.; Taparugssanagorn, A.; Tangparitkul, S.; Promsuk, N. Enhancing Short-Term Solar Photovoltaic Power Forecasting Using a Hybrid Deep Learning Approach. *IEEE Access* **2024**, *12*, 108928–108941. [CrossRef]
52. Babalhavaeji, A.; Radmanesh, M.; Jalili, M.; Gonzalez, S.A. Photovoltaic generation forecasting using convolutional and recurrent neural networks. *Energy Rep.* **2023**, *9*, 119–123. [CrossRef]
53. Lim, S.C.; Huh, J.H.; Hong, S.H.; Park, C.Y.; Kim, J.C. Solar Power Forecasting Using CNN-LSTM Hybrid Model. *Energies* **2022**, *15*, 8233. [CrossRef]
54. Limouni, T.; Yaagoubi, R.; Bouziane, K.; Guissi, K.; Baali, E.H. Accurate one step and multistep forecasting of very short-term PV power using LSTM-TCN model. *Renew. Energy* **2023**, *205*, 1010–1024. [CrossRef]
55. Renold, A.P.; Sinha, N.; Gao, X.Z. Hybrid machine learning approach for improved short-term PV power forecasting accuracy. *Results Eng.* **2025**, *28*, 107374. [CrossRef]
56. Yuan, W.; Ding, J.; Zhang, L.; Ni, J.; Zhang, Q. Short-Term Probabilistic Prediction of Photovoltaic Power Based on Bidirectional Long Short-Term Memory with Temporal Convolutional Network. *Energies* **2025**, *18*, 5373. [CrossRef]
57. Seman, L.O.; Stefenon, S.F.; Yow, K.C.; Coelho, L.D.S.; Mariani, V.C. Multi-step short-term solar energy forecasting using Fourier-enhanced BiLSTM and neural additive models. *Renew. Energy* **2026**, *257*, 124738. [CrossRef]
58. Luo, X.; Zhang, D.; Zhu, X. Combining transfer learning and constrained long short-term memory for power generation forecasting of newly-constructed photovoltaic plants. *Renew. Energy* **2022**, *185*, 1062–1077. [CrossRef]
59. Santos, M.L.; García-Santiago, X.; Camarero, F.E.; Gil, G.B.; Ortega, P.C. Application of Temporal Fusion Transformer for Day-Ahead PV Power Forecasting. *Energies* **2022**, *15*, 5232. [CrossRef]
60. Stefanov, D.; Demšar, J. Forecasting solar power production by using satellite images. *Sol. Energy 300*, 113823. [CrossRef]
61. Mazen, F.M.A. Transformer Model and DILATE Loss Function. *Energies* **2023**, *16*, 8105. [CrossRef]
62. Liao, Z.; Min, W.; Li, C.; Wang, B. Photovoltaic Power Prediction Based on Irradiation Interval Distribution and Transformer-LSTM. *Energies* **2024**, *17*, 2969. [CrossRef]
63. Dai, Y.; Wang, Y.; Chen, Y.; Wu, J.; Chao, J. Combining meteorological and power information of station-measurement and model-prediction with the hybrid CNN-Transformer and CNN-BiLSTM for ultra-short-term photovoltaic power forecasting. *Int. J. Electr. Power Energy Syst.* **2025**, *171*, 111009. [CrossRef]
64. Xu, S.; Ma, H.; Ekanayake, C.; Cui, Y. Swin transformer-based transferable PV forecasting for new PV sites with insufficient PV generation data. *Renew. Energy* **2025**, *246*, 122824. [CrossRef]
65. Qu, J. Day-ahead hourly photovoltaic power forecasting using attention-based CNN-LSTM neural network embedded with multiple relevant and target variables prediction pattern. *Energy* **2021**, *232*, 120996. [CrossRef]
66. Zhou, H.; Zhang, Y.; Yang, L.; Liu, Q.; Yan, K.; Du, Y. Short-Term photovoltaic power forecasting based on long short term memory neural network and attention mechanism. *IEEE Access* **2019**, *7*, 78063–78074. [CrossRef]
67. Zhu, H.; Wang, Y.; Sun, S.; Zhang, D.; Hu, S.; Chen, W. Meteorological information calculation and power forecasting for regional distributed photovoltaic power based on information fusion and deep learning. *Int. J. Electr. Power Energy Syst.* **2025**, *172*, 111189. [CrossRef]
68. Das, O.; Dahlioui, D.; Zafar, M.H.; Akhtar, N.; Moosavi, S.K.R.; Sanfilippo, F. Ultra-short term PV power forecasting under diverse environmental conditions: A case study of Norway. *Energy Convers. Manag. X* **2025**, *27*, 101072. [CrossRef]

69. Yu, D.; Choi, W.; Kim, M.; Liu, L. Forecasting day-ahead hourly photovoltaic power generation using convolutional self-attention based long short-term memory. *Energies* **2020**, *13*, 4017. [CrossRef]
70. Zang, H.; Cheng, L.; Ding, T.; Cheung, K.W.; Wei, Z.; Sun, G. Day-ahead photovoltaic power forecasting approach based on deep convolutional neural networks and meta learning. *Int. J. Electr. Power Energy Syst.* **2019**, *118*, 105790. [CrossRef]
71. Perera, M.; De Hoog, J.; Bandara, K.; Senanayake, D.; Halgamuge, S. Day-ahead regional solar power forecasting with hierarchical temporal convolutional neural networks using historical power generation and weather data. *Appl. Energy* **2024**, *361*, 122971. [CrossRef]
72. Huang, C.J.; Kuo, P.H. Multiple-Input Deep Convolutional Neural Network Model for Short-Term Photovoltaic Power Forecasting. *IEEE Access* **2019**, *7*, 74822–74834. [CrossRef]
73. Almaghrabi, S.; Rana, M.; Hamilton, M.; Rahaman, M.S. Multivariate solar power time series forecasting using multilevel data fusion and deep neural networks. *Inf. Fusion* **2024**, *104*, 102180. [CrossRef]
74. Michalakopoulos, V.; Spyropoulos, L.; Ilias, L.; Sarmas, E.; Marinakis, V.; Askounis, D. A comparative analysis of federated learning strategies for short-term photovoltaic power forecasting. *Energy Rep.* **2025**, *13*, 4223–4235. [CrossRef]
75. Moradzadeh, A.; Moayyed, H.; Mohammadi-Ivatloo, B.; Vale, Z.; Ramos, C.; Ghorbani, R. A novel cyber-Resilient solar power forecasting model based on secure federated deep learning and data visualization. *Renew. Energy* **2022**, *211*, 697–705. [CrossRef]
76. Riedel, P.; Belkilani, K.; Reichert, M.; Heilscher, G.; von Schwerin, R. Enhancing PV feed-in power forecasting through federated learning with differential privacy using LSTM and GRU. *Energy AI* **2024**, *18*, 100452. [CrossRef]
77. Zhang, W.; Chen, X.; He, K.; Chen, L.; Xu, L.; Wang, X.; Yang, S. Semi-asynchronous personalized federated learning for short-term photovoltaic power forecasting. *Digit. Commun. Netw.* **2023**, *9*, 1221–1229. [CrossRef]
78. Qu, Y.; Xu, J.; Sun, Y.; Liu, D. A temporal distributed hybrid deep learning model for day-ahead distributed PV power forecasting. *Appl. Energy* **2021**, *304*, 117704. [CrossRef]
79. Wang, Z.; Liu, H.; Wu, S.; Liu, N.; Liu, X.; Hu, Y.; Fu, Y. Explainable time-varying directional representations for photovoltaic power generation forecasting. *J. Clean. Prod.* **2024**, *468*, 143056. [CrossRef]
80. Xiong, B.; Chen, Y.; Zhao, X.; Su, Z.; Fu, J.; Chen, D.; Zhang, D. Multimodal ultra-short-term probabilistic solar power forecasting with generative AI and Transformer. *Adv. Appl. Energy* **2025**, *20*, 100250. [CrossRef]
81. Chen, D.; Shi, X.; Jiang, M.; Zhu, S.; Zhang, H.; Zhang, D.; Chen, Y.; Yan, J. Selecting effective NWP integration approaches for PV power forecasting with deep learning. *Sol. Energy* **2025**, *301*, 113939. [CrossRef]
82. Zhang, J.; Tan, Z.; Wei, Y. An adaptive hybrid model for day-ahead photovoltaic output power prediction. *J. Clean. Prod.* **2020**, *244*, 118858. [CrossRef]
83. Bai, R.; Li, J.; Liu, J.; Shi, Y.; He, S.; Wei, W. Day-ahead photovoltaic power generation forecasting with the HWGC-WPD-LSTM hybrid model assisted by wavelet packet decomposition and improved similar day method. *Eng. Sci. Technol. An. Int. J.* **2025**, *61*, 101889. [CrossRef]
84. Zang, H.; Cheng, L.; Ding, T.; Cheung, K.W.; Liang, Z.; Wei, Z.; Sun, G. Hybrid method for short-term photovoltaic power forecasting based on deep convolutional neural network. *IET Gener. Transm. Distrib.* **2018**, *12*, 4557–4567. [CrossRef]
85. Abed, J.N.; Abdoos, A.A.; Ebadi, A. A new hybrid intelligent method for short-term photovoltaic power forecasting based on combination of ELM, VMD and PCA. *Electr. Power Syst. Res.* **2026**, *250*, 112104. [CrossRef]
86. El Robrini, F.; Amrouche, B.; Cali, U.; Ustun, T.S. Assessment of machine and deep learning models integrated with variational mode decomposition for photovoltaic power forecasting using real-world data from the semi-arid region of Djelfa, Algeria. *Energy Convers. Manag. X* **2025**, *27*, 101108. [CrossRef]
87. Liu, L.; Zhang, J.; Xue, S. Photovoltaic power forecasting: Using wavelet threshold denoising combined with VMD. *Renew. Energy* **2025**, *249*, 123152. [CrossRef]
88. Fentis, A.; Rafik, M.; Bahatti, L.; Bouattane, O.; Mestari, M. Data driven approach to forecast the next day aggregate production of scattered small rooftop solar photovoltaic systems without meteorological parameters. *Energy Rep.* **2022**, *8*, 3221–3233. [CrossRef]
89. Li, P. A hybrid deep learning model for short-term PV power forecasting. *Appl. Energy* **2020**, *259*, 114216. [CrossRef]
90. Wang, H.; Yi, H.; Peng, J.; Wang, G.; Liu, Y.; Jiang, H.; Liu, W. Deterministic and probabilistic forecasting of photovoltaic power based on deep convolutional neural network. *Energy Convers. Manag.* **2017**, *153*, 409–422. [CrossRef]
91. Song, Z.; Xiao, F.; Chen, Z.; Madsen, H. Probabilistic ultra-short-term solar photovoltaic power forecasting using natural gradient boosting with attention-enhanced neural networks. *Energy AI* **2025**, *20*, 100496. [CrossRef]
92. Abumohsen, M.; Owda, A.Y.; Owda, M.; Abumihsan, A. Hybrid machine learning model combining of CNN-LSTM-RF for time series forecasting of Solar Power Generation. *E-Prime Adv. Electr. Eng. Electron. Energy* **2024**, *9*, 100636. [CrossRef]
93. Wang, Z.; Wang, C.; Cheng, L.; Li, G. An approach for day-ahead interval forecasting of photovoltaic power: A novel DCGAN and LSTM based quantile regression modeling method. *Energy Rep.* **2022**, *8*, 14020–14033. [CrossRef]
94. AlShafeey, M.; Csáki, C. Evaluating neural network and linear regression photovoltaic power forecasting models based on different input methods. *Energy Rep.* **2021**, *7*, 7601–7614. [CrossRef]

95. Theocharides, S.; Makrides, G.; Livera, A.; Theristis, M.; Kaimakis, P.; Georghiou, G.E. Day-ahead photovoltaic power production forecasting methodology based on machine learning and statistical post-processing. *Appl. Energy* **2020**, *268*, 115023. [CrossRef]
96. Renkema, Y.; Visser, L.; AlSkaif, T. Enhancing the reliability of probabilistic PV power forecasts using conformal prediction. *Sol. Energy Adv.* **2024**, *4*, 100059. [CrossRef]
97. Zhang, W.; Qiu, J. Day-ahead hybrid probability density prediction of photovoltaic power generation based on improved similar days and MCQRNN-GAQ. *Electr. Power Syst. Res.* **2026**, *252*, 112296. [CrossRef]
98. Yan, X.; Cao, Y. A three-stage hybrid model for short-term photovoltaic power prediction. *Sol. Energy* **2025**, *301*, 113904. [CrossRef]
99. Salameh, T.; Farag, M.M.; Hamid, A.K.; Hussein, M. Adaptive neuro-fuzzy inference system for accurate power forecasting for on-grid photovoltaic systems: A case study in Sharjah, UAE. *Energy Convers. Manag. X* **2025**, *26*, 100958. [CrossRef]
100. Liu, J.; Li, K.; Xue, W. Photovoltaic Solar Power Prediction Using iPSO-Based Data Clustering and AdaLSTM Network. *Energies* **2024**, *17*, 1624. [CrossRef]
101. Mayer, M.J.; Yang, D.; Markovics, D. The complexity and dimensionality of making deterministic photovoltaic power forecasts from ensemble numerical weather prediction. *Energy Convers. Manag.* **2025**, *344*, 120303. [CrossRef]
102. Salamanis, A.I.; Xanthopoulou, G.; Bezas, N.; Timplalexis, C.; Bintoudi, A.D.; Zyglakis, L.; Tsolakis, A.C.; Ioannidis, D.; Kehagias, D.; Tzovaras, D. Benchmark comparison of analytical, data-based and hybrid models for multi-step short-term photovoltaic power generation forecasting. *Energies* **2020**, *13*, 5978. [CrossRef]
103. Ogliari, E.; Dolara, A.; Manzolini, G.; Leva, S. Physical and hybrid methods comparison for the day ahead PV output power forecast. *Renew. Energy* **2017**, *113*, 11–21. [CrossRef]
104. Yamamoto, H.; Kure, T.; Kondoh, J.; Kodaira, D. Output Forecasting for Multiple Geographically Distributed PVs Without Meteorological Data. *IEEE Access* **2024**, *12*, 86997–87013. [CrossRef]
105. Pereira, S.; Canhoto, P.; Oozeki, T.; Salgado, R. Comprehensive approach to photovoltaic power forecasting using numerical weather prediction data and physics-based models and data-driven techniques. *Renew. Energy* **2025**, *251*, 123495. [CrossRef]
106. Luo, X. Deep learning based forecasting of photovoltaic power generation by incorporating domain knowledge. *Energy* **2021**, *225*, 120240. [CrossRef]
107. Dai, Y.; Wang, Y.; Leng, M.; Yang, X.; Zhou, Q. LOWESS smoothing and Random Forest based GRU model: A short-term photovoltaic power generation forecasting method. *Energy* **2022**, *256*, 124661. [CrossRef]
108. Sagingalieva, A.; Komornyik, S.; Senokosov, A.; Joshi, A.; Mansell, C.; Tsurkan, O.; Pinto, K.; Pflitsch, M.; Melnikov, A. Photovoltaic power forecasting using quantum machine learning. *Sol. Energy* **2025**, *302*, 114016. [CrossRef]
109. Yan, K.; Liu, J.; Zhang, J.; Yang, F.; Gao, Y.; Du, Y. Robust photovoltaic forecasting under severe data missingness via multi-domain collaboration and covariate interaction. *Appl. Energy* **2025**, *401*, 126771. [CrossRef]
110. Atiea, M.A.; Abdelghaffar, A.A.; Aribia, H.B.; Noureddine, F.; Shaheen, A.M. Photovoltaic power generation forecasting with Bayesian optimization and stacked ensemble learning. *Results Eng.* **2025**, *26*, 104950. [CrossRef]
111. Sarmas, E.; Spiliotis, E.; Stamatopoulos, E.; Marinakis, V.; Doukas, H. Short-term photovoltaic power forecasting using meta-learning and numerical weather prediction independent Long Short-Term Memory models. *Renew. Energy* **2023**, *216*, 118997. [CrossRef]
112. Khan, W.; Walker, S.; Zeiler, W. Improved solar photovoltaic energy generation forecast using deep learning-based ensemble stacking approach. *Energy* **2022**, *240*, 122812. [CrossRef]
113. Cai, C.; Liu, H.; Duan, Z. Dynamic stacking ensemble hybrid model for enhanced short-term photovoltaic power forecasting with self-organizing maps and advanced deep learning. *Energy Rep.* **2025**, *13*, 4280–4298. [CrossRef]
114. Tania, F.I.; Rani, P.; Ahmed, T.; Ahmad, S. A stacked generalization approach for day ahead hourly photovoltaic power forecasting. *E-Prime-Adv. Electr. Eng. Electron. Energy* **2025**, *12*, 100977. [CrossRef]
115. Mohammed, A.A.; Aung, Z. Ensemble learning approach for probabilistic forecasting of solar power generation. *Energies* **2016**, *9*, 1017. [CrossRef]
116. Rana, M.; Koprinska, I.; Agelidis, V.G. Univariate and multivariate methods for very short-term solar photovoltaic power forecasting. *Energy Convers. Manag.* **2016**, *121*, 380–390. [CrossRef]
117. Mei, F.; Gu, J.; Lu, J.; Lu, J.; Zhang, J.; Jiang, Y.; Shi, T.; Zheng, J. Day-Ahead nonparametric probabilistic forecasting of photovoltaic power generation based on the LSTM-QRA ensemble model. *IEEE Access* **2020**, *8*, 166138–166149. [CrossRef]
118. Liu, L. Prediction of short-term PV power output and uncertainty analysis. *Appl. Energy* **2018**, *228*, 700–711. [CrossRef]
119. Moreira, M.O.; Kaizer, B.M.; Ohishi, T.; Bonatto, B.D.; de Souza, A.C.Z.; Balestrassi, P.P. Multivariate Strategy Using Artificial Neural Networks for Seasonal Photovoltaic Generation Forecasting. *Energies* **2023**, *16*, 369. [CrossRef]

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